

IKS DEFINITION (INDIAN KNOWLEDGE SYSTEM)

Systematic, well-structured system and process of knowledge transfer from generation to generation.

The term 'Indian Knowledge System' (IKS) refers to the cumulative knowledge, wisdom, and traditions that have developed within the Indian subcontinent over centuries.

Breaking it down:

"**Indian**" denotes the geographical and cultural context of the knowledge, highlighting its origin within the undivided Indian subcontinent.

"**Knowledge**" encompasses a wide range of domains, including science, philosophy, arts and more.

"**System**" suggests that IKS is an organized, structured, and interconnected body of knowledge that is comprehensive

HISTORICAL PERSPECTIVE OF INDIAN CIVILIZATION 3000BCE TO 2000CE

1. **ANCIENT PERIOD**- stone age to – 7 th century(3300BCE -700CE)

2. **MEDIEVAL** – 8th – 18 th century (700-1857 CE)

- Early medieval - 8th – 13 th century

- Late medieval 13th – 18 th century

3.**MODERN PERIOD** -18 th century onward

- Colonial period (1757-1947)

- Mughal Empire (1526 – 1857 CE):

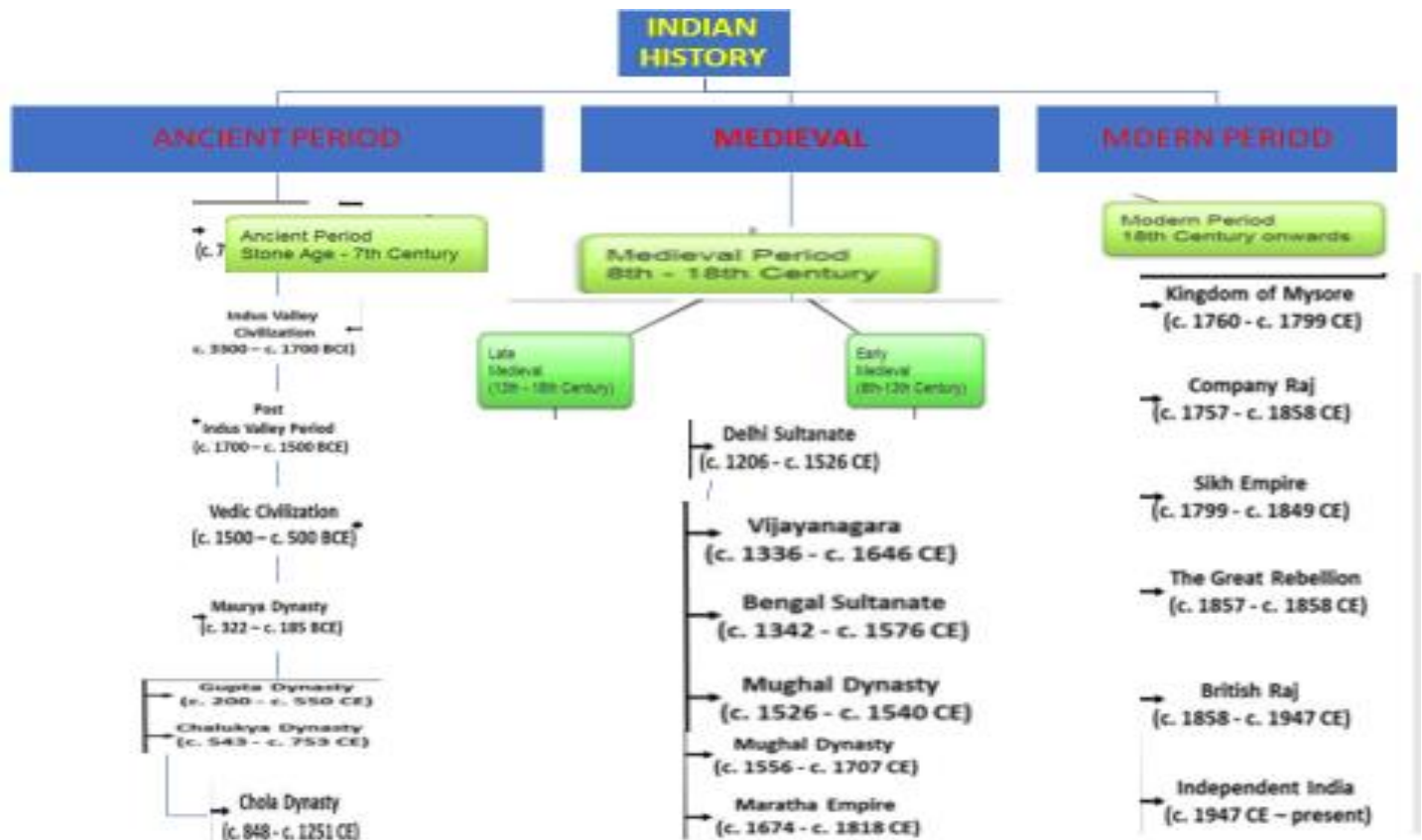
- British Colonial Rule (1858 – 1947 CE)

- Portuguese-rule (1505-1961)

- Indian Independence and Partition (1947):

- Post-Independence India

- Republic era-January 26,1950



• **Indus Valley Civilization Ancient India (Pre-history to 700 CE):**

- **Paleolithic Age (2 million BC – 10,000 BC):** Early humans used simple stone tools.

- **Mesolithic Age (10,000 – 8,000 BC):** Development of microliths (small stone tools).

- **Neolithic Age (8000 BC – 4,000 BC):** Agriculture and domestication of animals emerged.

- **Chalcolithic Age (4000 BC – 1500 BC):** First use of copper.

- Key Eras of Ancient India

- Prehistoric Era (Up to 3300 BCE): This period includes the Stone Age (Paleolithic, Mesolithic, and Neolithic). Notable for the Bhimbetka Rock Shelters (one of the oldest cave paintings) and the transition to settled agriculture in sites like Mehrgarh.

- Indus Valley Civilisation (c. 3300 – 1300 BCE): Also known as the Harappan Civilisation, it was one of the world's first urban cultures. It is famous for its advanced city planning, baked brick houses, and sophisticated drainage systems in cities like Mohenjo-daro and Harappa.
- Vedic Period (c. 1500 – 600 BCE): This age saw the arrival of the Indo-Aryans and the composition of the Vedas, the foundational texts of Hinduism. It marked the shift from a pastoral society to a more structured, agricultural one, and the emergence of the Varna (caste) system.
- Mahajanapadas & New Religions (c. 600 – 321 BCE): A "second urbanisation" occurred with the rise of 16 great kingdoms (Mahajanapadas). This was a time of intellectual flux, leading to the birth of Buddhism and Jainism as challenges to Vedic orthodoxy.
- Mauryan Empire (322 – 185 BCE): Founded by Chandragupta Maurya, it was the first empire to unify almost the entire Indian subcontinent. Its most famous ruler, Ashoka the Great, is renowned for embracing and spreading Buddhism following the bloody Kalinga War.
- Classical Period / Golden Age (320 – 550 CE): Under the Gupta Empire, India experienced a peak in science, art, and literature. Major achievements included the discovery of zero by mathematicians like Aryabhata and the literary masterpieces of Kalidasa.

Medieval India (700 CE – 1857 CE):

- **Early Medieval Period (700-1200 CE):** Regional kingdoms, including the Cholas in the south, Rajputs in the north, and others, flourished.
- **Delhi Sultanate (1206 – 1526 CE):** Five dynasties of Turkic and Afghan origin ruled over Delhi and much of India.
- **Mughal Empire (1526 – 1857 CE):** Founded by Babur, reaching peak under Akbar. A powerful empire known for its grand architecture (Taj Mahal), art, and culture.

Modern India (1857 – Present):

- **British Colonial Rule (1757 – 1947 CE):**
- The British East India Company established control, later succeeded by British Raj. The British East India Company gradually gained control, followed by direct rule from the British Crown.
- **Indian Independence and Partition (1947):**
- India gained independence from British rule, but the subcontinent was also partitioned into India and Pakistan, leading to mass migrations and violence.
- **Post-Independence India:**
- India became a republic 26 January 1950, adopted a democratic system, and has since experienced significant economic and social development
- Economic shift and liberalization (1980s 1990s)
- Science and technology 20th century
- Nuclear status 1998

EDUCATION SYSTEM IN ANCIENT INDIA

According to the eras, our education is divided into different systems:

1. Ancient Education or Gurukul Education System in Vedic era (3000 B.C.E) and Buddhist era (500 B.C. to 1200 CE)
2. Medieval Education System Or Islamic Education System in Muslim era (1200 CE. to 1700 CE)
3. British Education System Or Pre-Independence Education System in British era (1800 CE. to 1947 CE)
4. Post Independence Education System (After 1947)
5. Modern Education System



Figure 2: Classifications of Indian Knowledge Systems

According to the eras, Our education is divided into different systems:

1. Ancient Education or Gurukul Education System in Vedic era (3000 B.C.E) and Buddhist era (500 B.C. to 1200 CE)
 - Vedic period: Beginnings of education with oral traditions in gurukuls.
 - Ancient universities: Nalanda and Taxila flourish as centers of learning.
2. Medieval Education System Or Islamic Education System in Muslim era (1200 CE. to 1700 CE)
 - Medieval era: Islamic influence on education; Persian and Arabic teachings.
3. British Education System Or Pre-Independence Education System in British era (1800 CE. to 1947 CE)
 - Colonial rule: Introduction of British education system; emphasis on English
4. Post Independence Education System (After 1947)
 - 1947: India gains independence, focuses on literacy and elementary education.
5. Modern Education System
 - Modern era: Emergence of IITs, IIMs; technology-driven education revolution..

1. ANCIENT EDUCATION

Ancient education system evolved over the period and focused on the holistic development of the individual by taking care of both the inner and the outer self.

During the ancient period, two systems of education were developed, Vedic, and Buddhist. The medium of language during the Vedic system was Sanskrit, while those in the Buddhist system were Pali.

During those times the education was of Vedas, Brahmanas, Upanishads, and Dharmasutras. From the Rigveda onwards, our ancient education started with the objective of developing the students not only in the outer body but also on the inner body.

The ancient education focused on imparting ethics like humility, truthfulness, discipline, self-reliance, and respecting all creations to the students. The education was mostly imparted in ashrams, gurukuls, temples, houses. Sometimes pujaris of the temples used to teach students.

SOURCES OF EDUCATION

- The ancient system of education was the education of the Vedas, Brahmanas, Upanishads and Dharmasutras

AIM OF EDUCATION

- The main objective of education was to equip the students with a good quality of education.
- The education mostly focused on the enrichment of culture, character, and personality, development, and cultivation of noble ideals.
- The objective was gaining the mental, physical, and intellectual personality of students, to make the students future-ready and survive in any situation.
- The ancient education focused on imparting ethics like humility, truthfulness, discipline, self-reliance, and respecting all creations to the students.

CHARACTERISTICS OF ANCIENT EDUCATION

- During the ancient period, the state government and the people did not interfere in designing curriculum, payments of fees, regulation of teaching hours.
- There was a strong bonding between teacher and student.
- Every student was allotted with one teacher and more emphasis was given to the student-teacher relationship, each student used to meet teachers personally to learn and gain instructions from them.
- During ancient times, royal families, as well as kings of states, used to donate their wealth to improve the education system and quality.

CURRICULUM

- **It was dynamic and not static; it was made up of different stages.**
- The fundamental goal of building a good curriculum was to develop students physically and mentally.
- The curriculum consists of four Vedas, six vedangas, Upanishads, darshanas, Puranas, Tarka Shastra.
- Algebra, Geometry, and grammar were also given more importance at that time. Panini was famous in the domain of grammar at that time.
- The curriculum of the Buddhist system consists of pitakas, Abhidharma, and sutras. Besides this medicine, Vedas were also given importance.
- Hindu learning was a part of Buddhist learning, although more emphasis was given to Buddhist learning.
- Both the systems were going hand in hand at that time.
- The education was totally through orals and debates, and the exams were conducted every year.
- The education system of the ancient period focused on subjects like warfare, military, politics, religion.

2. Medieval education

During the eighth century Anno Domini (A.D) a huge number of Mohammadian invaded India. The Arabs and the Turks brought some new cultures, traditions, and institutions in India, in that the most remarkable change was the Islamic pattern of education which was different from the Buddhist and Brahmanic education system. The medieval age, education system primarily focused on the Islamic and Mughal System.

3. British education

In the 1830s Lord Thomas Babington Macaulay introduced the English language. The subjects and the syllabus were limited to some extent.

4. Post-Independence

After gaining independence in 1947, India's education history took a significant leap forward. The government implemented policies aimed at increasing access to education for all, emphasizing the importance of literacy and elementary education. The establishment of premier institutions like the Indian Institutes of Technology (IITs) and Indian Institutes of Management (IIMs) underscored India's commitment to fostering excellence in higher education.

5. Modern

As India entered the 21st century, the history of education continued to evolve. As time passed education started to develop and entered into the modern era that is in the twenty-first century, the era of science, technology, and innovations. Demand increases our education sector also needs to change and adapt to that environment.

The nation confronted challenges related to quality, accessibility, and equity in education. Yet, advancements in technology and the rise of e-learning platforms opened new horizons for learning. Online education, digital classrooms, and skill development programs have revolutionized knowledge dissemination, ushering in a new era of education in India.

Education system

1. **Gurukula system**
2. **Vishwavidyalayas; Ancient Indian Universities**

Gurukul System

The Gurukul was India's first system of education. The term "Gurukul" or "Gurukulam" is made from two different words – The Gurukul was India's first system of education.

The term “Gurukul” or “Gurukulam” is made from two different words – ‘Guru’ which means Teacher and ‘Kul’ which means family or home.

So basically, the Gurukul was actually the Guru or Acharya’s home where the students, also referred to as “Shishya” resided till they completed their education

It was a residential schooling system dating back to around 5000 BC, where shisya (student) and guru (teacher) used to reside in the guru's ashram (home) or in close proximity.

The Gurukul education system prepares the student for facing life and not only for a job. Character traits and holistic all-round development was encouraged. Value-based experimental education was imparted to the students.

The education was practically based and multiple disciplines were taught to the students like Languages, Yoga, Veda, Ayurveda, History & Politics, vocational training, self-defence, military combats etc. The students were taught to fulfill their duties towards self, family and society.

OBJECTIVES

1. Self Control
2. Development of character
3. Social Awareness
4. Integral development of personality
5. Propagation of purity
6. Preservation of knowledge and culture

The students were taught to fulfill their duties towards self, family and society.

1. Śikshā (Phonetics)
2. Vyākaraṇa (Grammar)
3. Jyotiṣa (Astronomy)
4. Arthaśāstra (Economics)
5. Dharmaśāstra (Laws)
6. Śāstravidyā (Art of Warfare)
7. Kālā (Fine Arts)
8. Mathematics,
9. fundamentals of Astronomy,
10. Science,
11. Languages
12. Medicinal theory

Vishwavidyalayas; Ancient Indian Universities

universities were known for their education, which included religious studies, medicine, literature, and more. .

Aims of Each Institute was

- To impart higher education
- To provide learnings of scriptures and ancient books
- To facilitate the comprehension of the content of the Vedas and the sacred books
- To impart education of arts and silpas (crafts)

Major universities was

1. Takshashila University
2. Nalanda University
3. Vikramshila University
4. Valabhi University
5. Somapura Mahavihara
6. Kanchipuram University
7. Pushpagiri University
8. Sharada Peethansta
9. Jagaddala Mahavihara
10. Odantapuri Mahavihara

Nalanda University

The word **NALANDA** derived from ‘nalam’ means lotus, (the symbol of knowledge) and ‘da’ means to give, so Nalanda means “Giver of Knowledge”. Nalanda (425BCE to 1205CE)

Location;- city of Rajagriha (now Rajgir) and about 90 kilometers (56 mi) southeast of Pataliputra now Patna). District Nalanda, Bihar The first residential university of the world. Nalanda an ancient centre for higher education .It was founded by the Kumaragupta (Gupta Empire), in 427CE and flourished for over 800 years till the end in 12th century CE

the first residential university of the world with capacity of 10000 students and professors, of these 1510 were teacher and 8500 students belonging to various level of attainment and studying various subjects, students from all over the world (Korea, Japan, China, Tibet, Indonesia, Persia, and Turkey) .

. Library at Nalanda Vidyapith

- Three buildings:

- (1) Ratnasagar,
- (2) Ratnadadhi and
- (3) Ratnaranjak

Hundred thousand Books on Religions, literature, astrology, astronomy and medicine available
It was devoted to Buddhist studies but also trained in

Courses taught in the University :

1. Political Science,
2. Administration,
3. Medicine,
4. Astronomy,
5. Mathematics,
6. Agriculture,
7. Trade,
8. Military Training,
9. Weapons Manufacturing,
10. Fine Arts,
11. Economics,
12. Philosophy,
13. Law

University is destroyed three times by kings at that times and also re- built twice by king Harshavardhan . But real destruction of nalanda university is done by Mohammad BakhtiyarKhilji in 1197.Efforts are being made to re-establish the Nalanda University by the Govt. of India

Takshashila University

Takshashila was in the kingdom of Gandhar, in Ancient India before partition.But now Takshashila is in Rawalpindi district of the Punjab Pakistan . Coordinates: 33°44'45"N 72°47'15"E

Takshashila nomenclature was derived from King Taksha –the nephew of Rama and son of Bharat , Taxila means city of cut-stoners

It was the first international university

Takshashila is the oldest among the universities in ancient India. It was well known as a centre of learning as early as during 700 B.C.

It had 10,500 students including those from Babylon, Greece, Syria, and China. 68 subjects were taught at this university and the minimum entry age, ancient texts show, was 16.A wide range of subjects were taught by experienced masters:

It is thought to have existed around the 5th century BC.

At this location, Chanakya is said to have written the Arthashastra.

Theologies of both Buddhism and Hinduism were taught here.

Takshashila's notable teachers and students include Chanakya, Charaka, Panini, Jivaka, Prasenajit, and others.

Time Period: While its exact founding date is debated, evidence suggests it dates back at least to the 5th century BCE and flourished for several centuries until its destruction around the 5th century CE

Curriculum: It offered a comprehensive curriculum, including religious teachings (particularly Buddhism), ancient scriptures, law, medicine, astronomy, military science, and the eighteen silpas or arts. Subjects like Vedas, Vedanta, Vyakaran, Ayurveda, and Surgery were also taught.

68 subjects were taught at this university and the minimum entry age, ancient texts show, was 16.

A wide range of subjects were taught by experienced masters:

1. Vedas,
2. Language,
3. Grammar,
4. Philosophy,
5. Medicine,
6. Surgery,
7. Archery,
8. Politics,
9. Warfare,
10. Astronomy,
11. Astrology,
12. Accounts,
13. Commerce,
14. Futurology,(study of the future)
15. Documentation,
16. Occult, (magical powers and activities)
17. Music,
18. Dance,

Vikramashila

It is a Buddhist tantric centre of learning, situated in **Kahalgao, District-** Bhagalpur, Bihar, India. Largest Buddhist university (Pala Dynasty)The Vikramasila Vihara (Buddhist monastery) was a famous seat of learning situated on a hillock on the banks of the Ganga in northern Magadha (Bihar) It was founded by king Dharmapala in the eighth century A.D. He gave liberal funds to the university so as to provide for free boarding and lodging to resident and non-resident monks studying at the place.It is said that the Vikramshila campus had six different buildings each belonging to a single stream of education, very much like we have different buildings for different departments now.The rage included topics like Sanskrit grammar, logic, metaphysics, philosophy, Buddhism, Tantra, and ritualism

Vikramshila University drew students from all over the country as well as from other countries. Over 1000 students were taught by over 100 teachers.

Subjects were taught here:

1. Political Science,
2. Administration,
3. Medicine,
4. Astronomy,
5. Mathematics,
6. Agriculture,
7. Trade,
8. Military Training,
9. Weapons Manufacturing,
10. Fine Arts,
11. Economics,
12. Philosophy,
13. Law

Valabhi University

Valabhi University is an ancient learning institution, located in modern-day Bhavnagar, Gujarat.

Valabhi University was a major Buddhist study center between 600 and 1400 CE. It was an important centre of Buddhist learning, and championed the cause of Hinayana Buddhism

This university used to receive royal funds for its sustenance and it was the lineage of the Maitrak dynasty, who later on became the patron of this university and designed its infrastructure further.

By the mid-Seventh century CE, Vallabhi had gained popularity for teaching Buddhist philosophy and Vedic literature.

It had more than 6000 students and 400 teachers.

Curriculum

- Imparted in subject's like
 1. Niti (Political science. Statesmanship),
 2. Varta (Business, Agriculture),
 3. Administration.
 4. Theology,
 5. Law,
 6. Economics and
 7. Accountancy.
- Education of 18 various subjects including four Vedas, six Vedangs, Puranas, Law, Epistemology, etc.
 - Jain philosophy and Baudhh philosophy were also added after some time in curriculum in which 72 different subjects were taught which include – reading, writing, calculation, grammar, astrology, chemistry, logic, medical science, history, music, painting, magic, hypnotism, trade, theatre, Sanskrit, etc.
 - Teachers at Valabhi include names like Sthirmati, Gunmati, Shri Mallavadi Suri (Jainacharya), Skandswami (Expert of Rigveda) and Nagarjuna
 - Valabhi was known for freedom and liberty for all people and it was also seen in its administration of education services.
 - Students were coming from all over the country to learn various subjects. They live at the vidyapith and learn by debates and discussions with the teachers and other scholars.

This university continued to function normally till the mid-Twelth century and was later destroyed by the Muslim invaders.

The foreign attacks had weakened the influence of local kings and it also disrupted the financial support this institution used to receive.

KNOWLEDGE OF MATERIALS AND PROCESSES:

Materials are closely connected to our culture. The development and advances of our society are depended on the available material and their uses. Early civilization designated by level of material use

Ancient civilizations possessed sophisticated knowledge of materials and processes, utilizing a wide range of natural resources to create tools, structures, art, and everyday objects. particularly in areas like metallurgy, construction, and agriculture

This knowledge, often acquired through generations of observation, experimentation, and skilled craftsmanship, is evident in various forms of archaeological evidence and historical texts.



Building Materials:

- **Diverse Materials:** Ancient Indian builders utilized a wide range of materials including mud, bricks (both sun-dried and fired), lime mortar, stone, and various types of wood.
- **Construction Materials:**
- **Stone:** Used extensively for walls, roofs, and floors due to its durability and availability in various types locally.
- **Bricks:** Manufactured from soil, these were a common building material, with techniques for soil selection, firing, and classification being well-established.
- **Mud:** Materials like mud, cob, and adobe blocks were widely used in construction, especially for walls and structures.
- **Wood:** Employed in construction, particularly for roofing and structural elements like floor joists and roof trusses, as seen in historical examples like rock-cut architecture.
- **Lime and Lime Mortars:** Limestone was excavated and burned to create lime, which was then used in mortars for construction.
- **Other natural materials:** Ash, cow dung, bark of various trees, bee wax, jaggery, rice husk, and various plant-based materials were also used as building components or binders in ancient Indian constructions.

Writing Materials:

Hard Materials: Materials like stone, metal, shells, and earthenware were used for writing, with techniques like engraving, embossing, painting, and scratching employed.

Soft Materials: Common writing surfaces included wooden boards, birch-bark (bhurja-patra), palm leaves (tala-patra), leather, and cotton cloths (karpasika pata).

Textiles and **Fabrics:** Woolen clothing, furs, and fine silk vestments

PERSONAL CARE:

Skincare & Cleansing (Ubtan): A mixture of turmeric, sandalwood, and gram flour (besan)

Body Oils (Abyanga): Daily massages with sesame, coconut, or almond oil were used for deep nourishment, supple skin, and improved circulation.

Hair Care: Hair was washed with natural, herbal concoctions of Reetha (soapberries), Amla (gooseberry), Hibiscus, and Shikakai.

Facial Care & Cosmetics: Sandalwood paste, turmeric, and milk masks were applied to the face. Kohl (collyrium) was used for eye makeup by both men and women.

Fragrance & Hygiene: Rose water, saffron, and other natural scents were used as perfumes and toners to keep the skin fragrant and healthy.

USE OF METAL

Ancient Indian have made use of base metals and alloys in multiple ways

1. Idols,
2. Utensils, vessels
3. Ornaments
4. Stamp seals
5. For the purposes of trade coins of Gold, Silver & Copper were in use.
6. Recorded history is 300-400 BCE However it could have been even earlier. Coins have inscriptions of the respective dynasties
7. Military aids (weapons)
8. Tools
9. Artefacts
10. Medical appliances
11. House and buildings
12. Ship building

(1) Gold (Suvarna): Symbolism, Religious Significance,, Economic Value:			(6) Artifacts,		
1. Jewellery,			7. sculptures,		
2. Religious artifacts,			8. tools, and weapons,		
3. Royal regalia.			9. Tamera jal		
4. Ornaments			10. Medicines		
5. Coins served as a medium of exchange and a store of value,			11. Utensils		
6. Facilitating trade and commerce			12. ALLOYING		
7. Clothing			(4) Iron (Loha): Symbolism, Religious Significance,, Economic Value:		
8. Construction.			1. Iron tools,		
9. ART AND PIGMENT			2. Weapons		
(2) Silver (Rajata): Symbolism, Religious Significance,, Economic Value:			3. Agricultural		
1. Jewelry,			4. Construction		
2. Religious artifacts,			5. MEDICINE		
3. Royal regalia.			6. Architecture		
4. Ornaments			(5) Steel (Ayas): Symbolism, Religious Significance,, Economic Value:		
5. Coins served as a medium of exchange and a store of value,			1. Iron tools,		
6. Facilitating trade and commerce			2. Weapons		
7. Clothing			3. Agricultural		
8. Construction.			4. Construction		
9. Crafting			5. Surgical Instruments		
10. Utensils,			(6) Alloys and Amalgams: Symbolism, Religious Significance,, Economic Value:		
(3) Copper (Tamra): Symbolism, Religious Significance,, Economic Value:			1. Sculptures,		
1. Jewellery,			2. Sacred idols		
2. Religious artifacts,			3. Temple bells.		
3. Ornaments			4. Construction		
4. Coins,			5. UTENSILS		
5. Construction.			6. Military and Defense		
			7. Medicinal (Rasayana)		

Material/ metal processing

Metallurgy is defined as a process that is used for the extraction of metals in their pure form. Ancient Indians contributions to metal working, metallurgy and metal science are noteworthy. India has had a high tradition of metallurgical skills. The two important sources for the history of Indian metallurgy are archaeological excavations and literary evidences.

The first evidence of metal in Indian subcontinent comes from Mehrgarh in Baluchistan, where a small copper bead was dated to about 6000 B.C.E

Archaeological excavations have shown that Harappan metalsmiths obtained copper ore from Aravalli Hills, Baluchistan or beyond. Many bronze figurines of humans and animals have been unearthed from Harappan sites

Systematic excavations at Mohenjodaro in Sindh and Harappa in Punjab show that during the mature Harappan period, the metal workers perfected the metallurgical skill. Harappans used metals like tin, arsenic, lead, antimony, etc., for alloying. They had also perfected the wax technique of metal casting in as early as the third millennium B.C.E. They melted and forged a variety of objects from metals such as lead, silver, gold and copper. They improved the hardness of copper for making artefacts by using tin and arsenic. These Vedas mention the extraction and processing of metals such as gold, silver, copper, tin, lead, iron and their alloys. Indian metalsmiths had notable achievements particularly developing unique skill of mining and working with iron, Steel, copper, Zinc and its alloys. India was the world leader in metallurgy for more than 12,000 years.

Historical Overview of Metallurgy in Ancient India

Pre-Indus Cultures (c. 7000–3300 BCE): Preceding the Indus Valley Civilization, several regional cultures in ancient India exhibited early signs of metalworking activities. Sites like Mehrgarh in present-day Pakistan have yielded copper artifacts dating back to the Neolithic period (c. 7000–2500 BCE), indicating the indigenous development of metallurgical knowledge and techniques. The discovery of copper smelting furnaces, crucibles, and metal objects

India was the world leader in metallurgy for more than 12,000 years

Indian metalsmiths had notable achievements particularly developing unique skill of mining and working with iron, Steel, copper, Zinc and its alloys

In-depth knowledge of ore-extraction, refining, alloying of copper, iron, tin, silver, gold, zinc, lead

In-depth knowledge of iron alloys which are resistant to corrosion and wear

List the important metal working process that was in practice 3000 years ago

(1) Early Beginnings: Metallurgical practices in ancient India can be traced back to the Indus Valley Civilization (c. 3300–1300 BCE), where evidence of copper smelting and metalworking has been discovered at sites such as Harappa and Mohenjo-Daro.

(2) Vedic Period and Iron Age: The Rigveda, one of the oldest sacred texts of Hinduism, contains references to metals such as gold, silver, copper, and iron, indicating the importance of metallurgical knowledge during the Vedic period (c. 1500–500 BCE).

(3) Mauryan and Gupta Empires: The Mauryan Empire (c. 322–185 BCE) and Gupta Empire (c. 320–550 CE) witnessed remarkable advancements in metallurgical technology.

The establishment of large-scale iron production centers, such as the famous iron pillar of Delhi commissioned by King Chandragupta II, exemplifies the metallurgical expertise of ancient Indian craftsmen.

The Gupta period also saw innovations in alloying techniques, resulting in the production of high-quality steel known as wootz steel.

Ancient Indians contributions to metal working, metallurgy and metal science are noteworthy.

1. Smelting-charcoal pit furnaces, bloomery furnaces
2. Casting -lost-wax casting (cire perdue), sand casting; shell moulding, clay moulding;
3. Forging, stamping (1200 bce)
4. welding
5. Alloying
6. Ore-extraction and ore-refining
7. Furnace and refractory's
8. Lost wax method,
9. Refractories & crucibles
10. Fuel
11. Temperature measurement
12. Refining-
13. Cupellation: the cupellation process was commonly used for refining noble metals like gold and silver
14. Quenching and tempering: for iron and steel production, refining processes included quenching and tempering

LIST THE IMPORTANT METAL WORKING PROCESS THAT WAS IN PRACTICE

techniques for Metal Extraction and Refinement

Mining Techniques:

(1) Surface Mining: Open-Pit Mining: Ancient Indians practiced open-pit mining to extract shallow deposits of metal ores, such as copper, iron, and lead. This method involved excavating large pits or trenches to access the ore bodies near the surface.

(2) Underground Mining: Shaft Mining: In regions with deeper ore deposits, ancient Indian miners utilized shaft mining techniques to access underground resources. Vertical shafts were dug into the earth, allowing miners to reach deeper layers where valuable metal ores were located.

(3) Alluvial Mining: Riverbed Mining: Alluvial deposits of metals, such as gold and silver, were often found in riverbeds and stream channels. Ancient Indians practiced alluvial mining by panning for gold and using simple sluice boxes to separate heavier metal particles from lighter sediments.

(4) Placer Mining: Beach Placer Mining: Along coastal areas, ancient Indian miners extracted placer deposits of heavy minerals, including ilmenite, rutile, and zircon, from sandy beaches and dune systems. Placer mining techniques involved hand tools and manual labor to collect and process beach sands.

Smelting Processes:

- **Charcoal Pit Furnaces:** Smelting of metal ores was typically conducted in charcoal pit furnaces, which were shallow, bowl-shaped structures dug into the ground. Ore and charcoal were layered inside these furnaces, and the fire was ignited at the bottom. The intense heat generated by burning charcoal facilitated the reduction of metal ores into molten metal.
- **Bloomery Furnaces:** Bloomery furnaces were used for smelting iron ore to produce wrought iron. These furnaces consisted of a clay or stone structure with a shaft for introducing ore and charcoal. Air was blown into the furnace through tuyeres to increase the temperature and facilitate the reduction of iron oxides.

Refining Techniques:

- **Cupellation:** The cupellation process was commonly used for refining noble metals like gold and silver. It involved heating the impure metal in a cupel—a porous container made of bone ash or clay. As the metal melted, base metals and impurities oxidized and were absorbed by the cupel, leaving behind purified gold or silver.
- **Quenching and Tempering:** For iron and steel production, refining processes included quenching and tempering. After smelting, the metal was heated to a critical temperature and then rapidly cooled (quenched) in water or oil to harden it. Subsequent tempering involved reheating the hardened metal to a specific temperature to improve its toughness and reduce brittleness.

Alloying Methods:

- **Pattern Welding:** Pattern welding was a technique used to create high-quality steel by forge-welding together layers of different iron and steel alloys. The layered structure increased the strength and flexibility of the resulting metal, making it suitable for manufacturing swords, knives, and other cutting tools.
- **Cementation:** Cementation involved packing wrought iron bars or sheets in powdered charcoal and heating them in a sealed container. Carbon from the charcoal diffused into the iron, transforming it into steel with improved hardness and strength.

Ancient Indians contributions to metal working , metallurgy and metal science are noteworthy.

Indian metalsmiths had notable achievements particularly developing unique skill of mining and working with iron , Steel , copper , Zinc and its alloys

India was the world leader in metallurgy for more than 12,000 years.

In-depth knowledge of ore-extraction, refining, alloying of copper, iron, tin, silver, gold, zinc, lead

In-depth knowledge of iron alloys which are resistant to corrosion and wear

TOOLS AND EQUIPMENT FOR METALWORKING TASKS

- (1) Forging Tools- including hammers, anvils, tongs, and swage blocks
2. Casting Implements-such as ladles, crucibles, and mould boxes
3. Cutting and Shaping Tools-including chisels, files, saws, and drills
4. Heat Treatment Equipment-use of charcoal-fired furnaces, quenching baths, and tempering ovens.
- (5) Measuring and Inspection Tools:- Ancient Indian metalworkers devised various measuring and inspection tools, including calipers, rulers, gauges, and templates, to ensure dimensional accuracy and quality control in metalworking processes.

MATHEMATICS:

Mathematics, referred to as Ganita is an integral part of Indians from very ancient times. Ancient Mathematics in India or the Vedic Mathematics had begun in the early Iron Age. Mathematics had played a significant role in the development of Indian culture and tradition

Mathematics, referred to as Ganita is an integral part of Indians from very ancient times. Ancient India made significant contributions to the field of mathematics

Ancient Mathematics in India or the Vedic Mathematics had begun in the early Iron Age. During the Indus valley civilization the sites of Harappa and Mohenjo-Daro constructions took place following precise mathematical calculation.

Many of the Vedic texts give details about some of the geometric constructions which were used during the Vedic Age.

Buddhist and Jain were also deeply interested in mathematical concepts and produced canonical works discussing some useful mathematical concepts.

Ancient Indians developed several concepts of mathematics primarily in the course of solving several real-life problems that they were facing.

History of Mathematics in India

- Vedic
- Classical
- Medieval to Mughal period
- Born in 1800s
- Born in 1900s



Chronology of Indian Mathematicians

- (c. 800 BCE): Author of the Baudhayana Sulba Sutra, containing early statements of the Pythagore **ANCIENT PERIOD (PRE-6TH CENTURY CE)**
- Baudhayana an theorem.
- Pingala (c. 3rd Century BCE): Described the binary system in his Chandaḥśāstra, linking long/short syllables to binary digits.
- Aryabhata (476–550 CE): Introduced place-value system, zero as placeholder, calculated pi (3.1416), and developed trigonometry in Aryabhatiya.
- **CLASSICAL PERIOD (400–1200 CE)**
- Brahmagupta (598–668 CE): Defined zero as a number, established rules for operations with zero and negatives, solved quadratic equations.
- Bhaskara I (c. 600–680 CE): Provided accurate sine tables and improved pi approximations.
- Mahavira (9th Century CE): Authored Ganita Sara Samgraha, worked on permutations, combinations, zero, and infinity.
- Bhaskara II (1114–1185 CE): Wrote Siddhanta Shiromani, worked on calculus concepts (differentiation, integration) and indeterminate equations.
- **MEDIEVAL PERIOD (1200–1800 CE)**
- Madhava of Sangamagrama (c. 1340–1425 CE): Founder of the Kerala School; developed infinite series for trigonometric functions, early calculus.
- nilakantha somayaji (1444–1544 CE): Advanced Kerala School work, proposed a planetary model similar to Copernicus.
- **Modern Period (1800–Present)**
- Srinivasa Ramanujan (1887–1920): Made major contributions to number theory, infinite series, continued fractions, and partitions.
- P.C. Mahalanobis (1893–1972): Statistician known for Mahalanobis distance and statistical analysis.
- C.R. Rao (1920–2023): Statistician known for Cramer-Rao bound and Rao-Blackwell theorem.
- Manjul Bhargava (b. 1974): Known for work in number theory, algebra, and geometry.

Early developments

- The Indus Valley civilization, which existed over 5,000 years ago, is considered the birthplace of Indian mathematics
- The Sulba Sutras from 800 BCE are the oldest known source of Indian mathematics
- The Shatapatha Brahmana and Taittiriya Samhita are Vedic mythology texts that describe altar constructions that show the origins of Indian geometry
- Indian mathematics emerged in the Indian subcontinent from 1200 BCE until the end of the 18th century.
- In the classical period of Indian mathematics (400 AD to 1200 AD), important contributions were made by scholars like Aryabhata, Brahmagupta, and Bhaskara II.
- The decimal number system in use today was first recorded in Indian mathematics.
- Indian mathematicians made early contributions to the study of the concept of zero as a number, negative numbers, arithmetic, and algebra.
- In addition, trigonometry was further advanced in India, and, in particular, the modern definitions of sine and cosine were developed there.
- These mathematical concepts were transmitted to the Middle East, China, and Europe and led to further developments that now form the foundations of many areas of mathematics

Vedic-

1. Baudhayana
2. Katyayana
3. Panini, ca. 5th c. BC, Algebraic grammarian
4. Yajnavalkya, credited with authorship of the Shatapatha Brahmana, which contains calculations related to altar construction.

Classical

- Post-Vedic Sanskrit to Pala period mathematicians (5th c. BC to 11th c. AD)
- 1. Aryabhata - Astronomer who gave accurate calculations for astronomical constants, 476AD-520AD
- 2. Aryabhata II
- 3. Bhaskara I
- 4. Brahmagupta - Helped bring the concept of zero into arithmetic (598 AD-670 AD)
- 5. Bhaskara II
- 6. Mahavira
- 7. Pavuluri Mallana - the first Telugu Mathematician
- 8. Varahamihira
- 9. Shridhara (between 650-850) - Gave a good rule for finding the volume of a sphere.

Medieval to Mughal period

- Narayana Pandit
- Madhava of Sangamagrama some elements of Calculus hi
- Parameshvara (1360–1455), discovered drk-ganita, a mode of astronomy based on observations, Madhava's Kerala school
- Nilakantha Somayaji, 1444-1545 - Mathematician and Astronomer, Madhava's Kerala school
- Mahendra Suri (14th century)

- Shankara Variyar (c. 1530)
- Raghunatha Siromani, (1475–1550), Logician, Navadvipa school

Born in 1800s Ramchundra (1821 – 1880)

He was British India's first major mathematician. His book, Treatise on Problems of Maxima and Minima, was promoted by the prominent mathematician Augustus

- Born in 1800s Ganesh Prasad (1876 – 1935)
He was an Indian mathematician who specialized in the theory of potentials, theory of functions of a real variable, Fourier series and the theory of surfaces..
- Born in 1800s Srinivāsa Rāmānujan (22 December 1887 – 26 April 1920)
He was an Indian mathematician and autodidact who, with almost no formal training in pure mathematics, made extraordinary contributions to mathematical analysis, number theory, infinite series and continued fractions. He was Independently compiled nearly 3900 results during his short lifetime.
- Born in 1800s Srinivāsa Rāmānujan (22 December 1887 – 26 April 1920)
He stated results that were both original and highly unconventional, such as the Ramanujan prime and the Ramanujan theta function, and these have inspired a vast amount of further research.
- Born in 1900s Harish-Chandra (11 October 1923 – 16 October 1983)
He was an Indian mathematician, who did fundamental work in representation theory, especially Harmonic analysis on semisimple Lie groups

USE OF MATHEMATICS

1. solving several real-life problems
2. Astronomy and calendrical calculations;- predictions of the movement of celestial bodies in the sky
3. Architecture and construction- The Sulbasutras, ancient Indian texts on geometry, provided precise instructions for building altars and other religious structures.
4. Taxation
5. Commerce
6. Trade
7. record keeping
8. Calendar
9. Structure
10. Jyotisha'
11. Surveying
12. Medicine
13. Art and music: Mathematical principles may have influenced various art forms, including the use of ratios and sequences,

CONTRIBUTION OF INDIAN MATHEMATICIANS

There is doubt that the world today is greatly indebted to the contributions made by Indian Mathematicians. One of the most important contribution made by Indian Mathematicians ARE

1. Introduction of decimal system
2. Invention of zero.
3. The number system
4. Vedic Mathematics and arithmetical operations • Geometry of the Sulba Sutras
5. Indeterminate Equations
6. Algebra
7. Indian Trigonometry
8. Infinite Series
9. Calculus.
10. Value of pi
11. Pithagoras theorem
12. Statistics:.
13. Number Theory
14. Combinatorics

AIRTHEMATIC

- Square of number
- Square root
- Square root of imperfect system
- Series and progression

GEOMETRY

- Property of right triangle
- The value of π

Trigonometry

Algebra

Binary mathematics and combinatorial problems

Magic square

Great mathematicians and their contribution

MATHEMATICS

TABLE 8.2 Ancient Indian Mathematicians and Their Salient Contributions

Sl. No.	Details of the Work/Mathematician	Period, Location	Salient Contributions
1	Vedic Texts	3000 BCE or earlier	The earliest recorded mathematical knowledge, number system, Pythagorean type triplets; Decimal system of naming numbers, the concept of infinity;
2	Lagadha – Vedāṅga-jyotiḥ	~ 1300 BCE	Astronomical concepts; a mathematical model for sun-moon movement in time; equinoxes and solstices;
3	Śulba-sūtras (Baudhāyana, Āpastamba, Kātyāyana and Mānava Śulba-sūtras)	800–600 BCE	Earliest Texts of Geometry; Approximate value of the square root of 2, and π ; Exact procedures for the construction and transformations of squares, rectangles, trapezia, etc.
4	Pāṇini – Aṣṭādhyāyī	500 BCE; Sālitura (in Khyber province in Pakistan)	Algorithmic approaches; Originator of the Backus-Naur Form (BNF), used in programming languages today; Context-sensitive rules, Arrays, inheritance, polymorphism;
5	Plāgala – Chandaḥ-śāstra	300 BCE	Binary sequences; Conversion of binary to decimal system and vice versa; 'Meru Prastara' (Pascal's triangle); Optimal Algorithms to calculate powers; Zero as a Symbol;
6	Bauddha Mathematical Works	about 500 BCE to 500 CE	Multi-valued logic, Discussion of indeterminate and infinite numbers;
7	Jaina Mathematical Works – Śūrya-prajñapti, Jambhūdvīpa-prajñapti, Bhāgavati-sūtra, Śākhadāyaka-sūtra, Uśādhāyaka-sūtra, Tīrthoṃśana-sūtra, Amogha-sūtra-sūtra	200 BCE to 300 CE	Concept of logarithms, large numbers; algorithms for raising a number by a power; the arc of a circle; combinatorics; mensuration; Decimal system; Approximation of π ;
8	Āryabhaṭa – Āryabhaṭīya	476–550 CE; Kusumapura, near Patliputra, Bihar	Concise verses; Algorithm for square root, cube root, Place value system; Sine table; geometry; quadratic equations; Linear indeterminate equations; Sums of squares and cubes of numbers; Planetary astronomy; Plane and spherical trigonometry;
9	Varāha Mihira – Brhat Samudra, Brhat-jātaka, Pañca-siddhāntikā	402–565 CE, Ujjain, Madhya Pradesh	Summary of five ancient siddhāntas; Sine table, trigonometric identities; ($\sin^2 + \cos^2$); combinatorics; Magic squares;
10	Bhāskara I – Commentary on Āryabhaṭīya, Lagha-bhāskariya and Mahā-bhāskariya	600–680 CE; Vallabhi region, Saurashtra, Gujarat	Expanded Āryabhaṭa's work on Integer solution for indeterminate equations; Approximate formula for the sine function, Planetary Astronomy;

Concept of Infinity, Decimal

Trigonometry

Binary System 0, 1

Algebra

Sl. No.	Details of the Work/Mathematician	Period, Location	Salient Contributions
11	Brahmagupta – Brahmasphuṭa-siddhānta; Khaṇḍakhadyaka	598–668 CE; Bhīllamala in Rajasthan	Rules of arithmetic operations with zero and negative numbers, Algebra (Bijaganita); linear and quadratic indeterminate equations; Pythagorean triplets, Formula for the diagonals and area of a cyclic quadrilateral; notion of arithmetic mean.
12	Virahāṅka – Vyttatātisamuccaya (in Prakṛt)	~600 CE	Fibonacci numbers; Moric metres.
13	Śrīdhara-cārya – Trisatīkā and Pāṭiganita	870–930 CE; Bhūrisṛṣṭi or Bhurshut village, Hugh, West Bengal	Arithmetic, Algebra, and Commercial Mathematics; Approximation of square root of a non-square number; Quadratic equations; Practical applications of algebra;
14	Mahāvīra-cārya – Gaṇita-sāra-saṅgraha	800–870 CE; Gulbarga Karnataka;	A comprehensive, exclusive textbook on mathematics covering arithmetic-geometry- algebra. Continuing the ancient jaina mathematics tradition; permutations and combinations; arithmetic and geometric series; the sum of squares and cubes of numbers in arithmetic progression;
15	Jayadeva	10th Century CE or earlier	Cakravāla (cyclic) method for solution of the second-order indeterminate equation.
16	Śrīpati – Gaṇita-tilaka, Siddhānta-śekhara, Dhikojidākarapa, etc.	1019–1066 CE; Rohiṇīkhanda, Maharashtra	Planetary Astronomy
17	Bhāskarā-cārya (Bhaskara-II) – Lilāvati on arithmetic and geometry; Bijaganita on algebra; Siddhānta-śiromaṇi on astronomy; Vāsanābhāṣya on Siddhānta-śiromaṇi.	1114–1185 CE; Hailed from Bijadavida	Canonical textbooks used all over India, Detailed explanations including Upapatti (demonstration or proof); addition formula for sine function. Surds; permutations, and combinations; Solution of indeterminate equations, Ideas of calculus, including mean value theorem, planetary astronomy; construction of several instruments;
18	Nārāyaṇa Paṇḍita – Gaṇita-kaumudī – a treatise on arithmetic and Bijaganita-aṭṭama – a treatise on algebra.	1325–1400 CE;	Advanced textbooks taking forward the works of Bhāskarā-cārya, further properties of cyclical quadrilaterals, summation and repeated summations of arithmetic series, theory and construction of Magic squares, further developments in combinatorics.

BAUDHAYANA was an Indian Mathematician who was born in 800 BC and died in 740 BC.

Baudhayana , also known as Baudhayana Acharya, is an ancient Indian mathematician and scholar known for his significant contributions to mathematics.

Bodhayana's work primarily focused on geometry and geometric principles.

Bodhayana elaborated on various mathematical topics, including arithmetic, algebra, and geometry.

He expanded upon the geometric concepts and theorems found in Brahmagupta's work, which included ideas related to:

1. Geometry: Bodhayana's commentary covered various aspects of geometry, including the properties of triangles, circles, and other geometric shapes. He explained the principles of constructing altars and geometric figures.
2. Algebra: Bodhayana contributed to the understanding of algebraic concepts and equations.

His commentary contained explanations of algebraic expressions and equations, demonstrating his mathematical prowess.

Baudhayana theorem;- A shloka from the Śulbasûtra is proof that he had the concept of Pythagoras theorem in his mind even before the Pythagoras was actually made

1. Shulabh sutra
2. Pre dated Pythagorean theorem
3. Square root
4. Approximation of $\sqrt{2}$
5. Area of triangle
6. Properties of Geometric Shapes
7. places.
8. Value of Pi : Approximated the value of Pi for circle-square conversions.
9. Circling a square.

Bodhayana even figured out:

Diagonals of a rhombus bisect at right angles,

Diagonals of a rectangle bisect each other,

The midpoints of a rectangle joined forms a rhombus whose area is half the rectangle, etc.

The area of a square formed by joining the middle points of a square is half of the original.

PINGALA

Pingala (circa 3rd–2nd century BCE)

Binary sequences; Conversion of Binary to Decimal system and vice versa; 'Meru Prastara' (Pascal's triangle); Optimal Algorithms to calculate powers; Zero as a Symbol;

Binary System Representation: He described a binary system where long (- guru) and short (l - laghu) syllables functioned as binary digits, using positional notation to count, which is a key forerunner to computer binary code.

Combinatorics and Algorithms: He developed algorithms (Prastara) to count all possible combinations of long and short syllables in a given meter, effectively utilizing combinatorial methods and binomial coefficients.

Fibonacci Numbers: In his study of matra-meru (meru-prastara), he identified the sequence now known as the Fibonacci sequence ((1,1,2,3,5,8,...)).

Binary-to-Decimal Conversion: He provided methods to calculate the (n) -th pattern in a binary series, effectively demonstrating a form of binary-to-decimal conversion.

Concepts of Zero: While not using a symbol for zero, his work indicated the early understanding of the concept of "empty" or void in positional notation.

VARAHAMIHIRA (505–587 CE)

Varahamihira (505–587 CE) was a pivotal figure in ancient Indian science whose mathematical work laid the foundation for advancements in trigonometry, combinatorics, and algebra.

Trigonometry: Improved the precision of Aryabhata's sine tables and established formulas such as

Algebraic Rules: Defined the properties of zero and negative numbers, including rules for adding/multiplying positive and negative numbers.

Combinatorics: Discovered a variant of Pascal's triangle used to compute binomial coefficients.

Magic Squares: Created the first known magic square.

Quadratic Equations: Provided a systematic treatment of algebraic equations in the Brihat-Samhita.

Mathematical Astronomy: In Pancha-Siddhantika, he applied these mathematical techniques to calculate planetary motions, eclipses, and the true value of the equinox.

Major Mathematical Works

Pancha-Siddhantika (The Five Treatises)- : Written around 575 CE, this is his most important work for mathematical astronomy. It summarizes five earlier (now lost) astronomical systems: Surya, Romaka, Paulisa, Vasishta, and Paitamaha

Brihat-Samhita- : An encyclopedic work containing 106 chapters. While primarily about divination, it covers a vast range of subjects including mathematics, architecture, and planetary motions.

Brihat Jataka- : Considered a foundational text for Indian astrology, it uses sophisticated mathematical calculations for horoscopy.

ARYABHATA (476—550AD)

Aryabhata was born in 476 A.D. Kusumpur, India. He was the first in the line of great mathematicians from the classical age of Indian Mathematics and Astronomy.

Aryabhata's Books:

Aryabhata has made a significant contribution to the field of Mathematics.

Aryabhata – Siddhanta- based on the astronomical computations done by the Aryabhata. This book contains information about the research he has done on stars, solar systems, planets, and the universe.

Aryabhata- This book covers mathematical theories. It includes topics like arithmetic, tables of sines, trigonometry, algebra, and fractions.

Key Mathematical Contributions:

1. •Decimal Number System:
2. Introduced the concept of zero and the place-value system.
3. Laid the foundation for the modern decimal system.
4. Approximation of Pi (π):
5. Calculated the value of Pi as 3.1416, remarkably close to the actual value.
6. Emphasized the importance of Pi in geometry and trigonometry.
7. Trigonometry:
8. Developed the sine (jya) and cosine (kojya) functions.
9. Provided tables of sines, which were later adopted in global mathematics.
10. Place Value System and Zero: Aryabhata is credited with the development of the place value system, and his works laid the foundation for the concept of zero.
11. Trigonometry: Aryabhata's work also included concepts in trigonometry, such as the sine and cosine functions

Key contributions include:

Place Value and Zero: He introduced the decimal system and used a placeholder for zero, enabling complex, large-number calculations.

Value of Pi (π): He calculated as 3.1416, remarkably close to the modern value, and identified it as an irrational number.

Trigonometry: He introduced the concepts of jya (sine) and kojya (cosine), creating one of the earliest sine tables based on the half-chord formula.

Algebra and Arithmetic: He developed the kuttaka method to solve indeterminate linear equations. He also provided formulas for the sum of the first n integers, as well as the sum of squares and cubes of the first n integers.

Geometry: He provided accurate formulas for the areas of triangles, circles, and trapeziums, as well as the volumes of spheres and pyramids.

Root Calculations: He formulated algorithms for calculating square and cube roots of integers



BRAHMAGUPTA (598-668 AD)

The most significant contribution of Brahmagupta was the introduction of zero(0) to the mathematics which stood for nothing. Brahmagupta was born in 598 A.D. in Bhinmal city in the state of Rajasthan. He was a mathematician and astronomer, who wrote many important works on mathematics and astronomy. His best known work is the "Brahmasphuta-siddhanta", written in 628 AD in Bhinmal.

• Arithmetic & Number System:

- Rules for Zero: First to define zero as a number in its own right, establishing that $a+0=a$, $a-0=a$, $a \times 0=0$, and defined $a/0$.

- Negative Numbers: Introduced the concept of negative numbers (debts) and positive numbers (fortunes), providing rules for their addition, subtraction, multiplication, and division.

• Algebra:

1. Quadratic Equations: Formulated methods to solve general quadratic equations $(ax^2 + bx = c)$ and gave both roots, including negative/irrational solutions.

1. Indeterminate Equations: Solved indeterminate quadratic equations, including a method known as the Bhavana for solving Pell's equation $(x^2 - ny^2 = 1)$.

2. Algorithms: Developed techniques for square and cube roots, and a multiplication method similar to the modern lattice method.

- Geometry & Trigonometry: Area = $\sqrt{(s-a)(s-b)(s-c)(s-d)}$, where s is the semi-perimeter.

- Brahmagupta's Formula: Discovered the formula for the area of a cyclic quadrilateral (four-sided figure inscribed in a circle):

- Brahmagupta's Theorem: Proved that in a cyclic quadrilateral with perpendicular diagonals, the perpendicular line from the intersection of diagonals to any side bisects the opposite side.

- Pythagorean Triples: Provided formulas to generate Pythagorean triples.

- Geometry & π : Approximated π as $\sqrt{10}$ and made contributions to trigonometry, including, sine tables.

- Series: Formulated rules to calculate the sum of series, including the sum of squares and cubes of the first natural numbers.

BHASKARACHARYA (1114-1185 AD)

He was born in Bijapur in modern Karnataka. He & his work represent a significant contribution to mathematical & astronomical knowledge in the 12th century. His main work "Siddhanta Shiromani" is divided into four parts called Lilawati, Bijaganit, Grahaganita and Goladhaya. These four sections deal with arithmetic, algebra, mathematics of planets and spheres respectively.

He was the first to give that any number divided by zero gives infinity.

He was written a lot about zero, surds, permutation and combination.

He wrote, "The hundredth part of the circumference of a circle seems to be straight. Our earth is a big sphere and that's why it appears to be flat."

He gave the formulae like: $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$.

He calculated derivatives of Trigonometric functions and formulae.

He developed spherical trigonometry along with a number of other trigonometric results.

He explained solution of quadratic, cubic and quartic indeterminate equations.

He developed a proof of Pythagoras Theorem by calculating the same area in two different ways & these cancel out terms to get $a^2 + b^2 = c^2$.

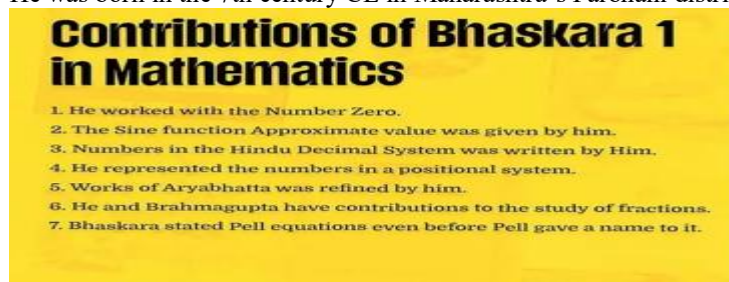
He gave first general method for finding the solution of the problem $x^2 - ny^2 = 1$ (so called Pell's equation).

He gave solution of Diophantine equations of second order such as $61x^2 + 1 = y^2$.



Bhaskara I

He was born in the 7th century CE in Maharashtra's Parbhani district.



Contributions of Bhaskara 1 in Mathematics

1. He worked with the Number Zero.
2. The Sine function Approximate value was given by him.
3. Numbers in the Hindu Decimal System was written by Him.
4. He represented the numbers in a positional system.
5. Works of Aryabhatta was refined by him.
6. He and Brahmagupta have contributions to the study of fractions.
7. Bhaskara stated Pell equations even before Pell gave a name to it.
8. He was the first to use the Brahmi numerals in a scientific contribution to Sanskrit.
9. He even contributed in prime numbers

Bhaskara II

- He was born in Bijapur, Karnataka in 1114 CE.
- He is celebrated as the greatest mathematician of the medieval period.
- He devised differential coefficient and differential calculus, prior to Newton and Leibniz.
- His famous work Siddhanta Shiromani has laid the base for various theories in arithmetic, algebra, etc.
- He has also written a treatise named Karana Kautuhala.

RAMANUJAN (1887-1920)

Ramanujan was born on 22nd of December 1887 in Erode, Madras Presidency. He made extraordinary contributions to mathematical analysis, number theory, infinite series, and continued fractions. He demonstrated unusual mathematical skill at school, winning accolades and awards.

By 17, he had conducted his own mathematical research on Bernoulli numbers and the Euler-Mascheroni constant.

He discovered theorems of his own and rediscovered Euler's identity independently.

He independently compiled nearly 3900 results (mostly identities and equations). Nearly all his claims have now been proved correct.

Ramanujan showed that any big number can be written as sum of not more than four prime numbers.

He showed that how to divide the number into two or more squares or cubes.

Ramanujan's Number: When Mr. G.H. Hardy came to see Ramanujan in taxi number 729, Ramanujan said that 1729 is the smallest number which can be written in the form of sum of cubes of two numbers in two ways, i.e. $1729 = 9^3 + 10^3 = 1^3 + 12^3$. Since then the number 1729 is called Ramanujan's number.

In 1918, Ramanujan and Hardy studied the partition function $P(n)$ extensively and gave a non-convergent asymptotic series that permits exact computation of the number of partition of an integer.

He discovered mock theta function in the last year of his life. For many years these functions were a mystery, but they are now known to be the holomorphic parts of harmonic weak mass forms.



Srinivasa Ramanujan (1887–1920)

Srinivasa Ramanujan is perhaps the most famous Indian mathematician of the modern era made extraordinary contributions to mathematical analysis, number theory, and infinite series.

- **Contributions:**
 - **Ramanujan's Notebooks:** His work, compiled in his notebooks, contains thousands of results, many of which were original and highly advanced.
 - **Partition Function:** Ramanujan developed a highly original approach to the partition function, which has been influential in the study of number theory.
 - **Mock Theta Functions:** His discovery of mock theta functions opened up new areas of research in mathematics.
- A self-taught mathematical genius who made extraordinary contributions to number theory, infinite series, and continued fractions.
- Discovered new identities and formulas related to elliptic curves and modular forms.
- Known for his work on the partition function and mock theta functions.
- Landau–Ramanujan constant
- Ramanujan conjecture
- Ramanujan theta function
- Ramanujan–Sato series
- Ramanujan prime
- Ramanujan theta function
- Ramanujan–Soldner constant
- Ramanujan's sum
- Ramanujan conjecture

C.R.Rao

- C.R. Rao is an Indian-American statistician and professor emeritus at Penn State University.
- He is known for pioneering work in multivariate analysis, statistical genetics, and DNA sequencing.
- He developed the Rao–Blackwell theorem and the Cramér–Rao bound.
- He was awarded the National Medal of Science in 2010. It is the highest scientific honor in the U.S.
- He was the president of the International Statistical Institute from 1985 to 1987.
- He was awarded the Padma Vibhushan in 2013 for his contributions to science.

SatyendraNath Bose

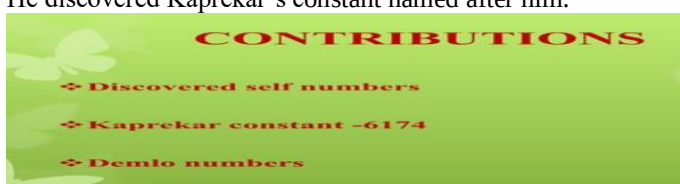
- He was born in Kolkata in the year 1894.
- He was given the prestigious Padma Vibhushan award.
- He was a theoretical physicist, laying the foundation for theoretical physics in India.
- He was known for his work on quantum mechanics, providing a base for Bose-Einstein statistics.
- Some of the other contributions of SatyendraNath Bose include Bose-Einstein condensate, Boson, Single field theory, Diffraction of X-ray and research in the field of Electromagnetic waves.

Shakuntala Devi

- She was born in 1929 in Karnataka's Bengaluru.
- She was a mental calculator commonly known as the "Human Computer".
- She competed against the world's then fastest computer, UNIVAC to calculate the 23rd root of a number with 201 digits and won by 10 seconds.
- She earned a place in The Guinness Book of World Records, 1982 edition.

D.R.Kaprekar

- DattatreyaRamchandraKaprekar was born in Dahanu in 1905.
- He was an Indian recreational mathematician describing multiple classes of natural numbers that included the Kaprekar, Harshad and self numbers.
- He discovered Kaprekar's constant named after him.



◦

Narendra Karmarkar

- Narendra Karmarkar is an Indian-American mathematician known for Karmarkar's algorithm. It is a polynomial time algorithm for linear programming problems.
- He developed this algorithm in 1984 while working at AT&T Bell Laboratories. It was a revolutionary improvement over the then-existing algorithms.
- Karmarkar's algorithm is exponentially faster than the simplex algorithm for large-scale linear programming problems.
- It laid the foundation for several new interior-point algorithms widely used today.
- He received the John von Neumann Theory Prize in 1989 for developing Karmarkar's algorithm.
- He retired from AT&T Bell Laboratories as the director of theoretical research in 1998.

ASTRONOMY

In past Astronomy was known as “Khagola-Shastra” or the Science of Cosmos (Khagola meaning Cosmos and Shastra meaning Science).

Khagolashastra ; -Astronomy, science that encompasses the study of all extraterrestrial objects and phenomena. Astronomy deals with – Planets, Stars and Galaxies

The study of the planetary motion and relative positions of celestial body.

A celestial body is a natural object outside of the Earth's atmosphere. This can include planets, sun, stars, moons, asteroids, comets, **Satellites**, or even nebulae and galaxies.

Astronomy is the study of space, celestial objects, and everything that isn't on Earth. This includes the study of the movement of stars, where the Earth lies in relation to other planets and celestial objects, and what makes up stars and other planets. Ancient astronomers' knowledge about planets wasn't as in-depth as our current knowledge, but they were able to differentiate many of the planets from the stars. In particular, they knew about the five planets that were visible to the naked eye: Mars, Venus, Saturn, Jupiter, and Mercury.

Indians learnt many elements of astronomy from the Greeks and the Romans due to closer contacts of India with Greco-Roman world during the post-Mauryan age. Astronomy was the only branch of knowledge in which Indians learnt from foreigners.

INDIAN ASTRONOMY HISTORY

Ancient Indian astronomy may be classified into two main

categories:

- **Ancient astronomy can be broadly classified into**
- **1. Vedic astronomy (c. 1500 – 500 BCE)**- The vedic astronomy is the astronomy of the vedic period i.e. the astronomy found in the vedic samhitas and brahmanas and allied literature. Vedic astronomy focused on timekeeping for rituals,
 - It focused on understanding the movements of the sun and moon to calculate days, months, seasons, and years.
- 1. Key aspects included the study of new moons, full moons, equinoxes, and solstices
- 2. **Post Vedic astronomy (c. 500 BCE – 1200 CE) - Siddhantic, astronomy and Zij astronomy**
 - Post-Vedic astronomy is the study of the Sun and Moon in ancient India, and the period between Vedic and siddhāntic astronomy
 - Siddhantic astronomy developed more sophisticated mathematical and astronomical models.
 - The term "Siddhanta" refers to astronomical texts containing solutions and explanations of celestial phenomena.
 - This era saw the development of more advanced mathematical and astronomical models, departing from the Vedic focus on timekeeping.
 - Key topics included the solar year, solstices, equinoxes, lunar periods, eclipses, and planetary movements.
 - Zij astronomy represents a later phase influenced by interaction with Arab astronomy.
 - The term "Zij" refers to astronomical handbooks or tables developed in the Arab world

This era represents the influence of Arab astronomy on Indian astronomy, particularly after the 8th century CE. This phase involved continued development and refinement of astronomical calculations and models

The Indus Valley Civilization, dating from 2500 BCE-1900 BCE, is one of the oldest civilizations in the world.

While much information about the study of astronomy during the period is not known because of the undeciphered Indus script,.

Evidence such as symbols on seals and pottery shards, has been interpreted by scholars to suggest that people were aware of the movements of the sun, moon, and stars, which likely played a role in their agricultural practices, religious beliefs, and societal organization.

Vedic period- The oldest known written records of astronomical concepts in the Indian subcontinent can be traced back to the Vedic period, dating from 1500-600 BCE.

The Vedic hymns indicate the knowledge of cosmological and astronomical concepts, used mostly for timekeeping, to perform sacrifices and rituals.

The texts were used to understand the movement of the sun, moon, stars, and other celestial bodies, along with solstices, seasons, equinoxes, and eclipses.

The texts also mention the way of calculating time and days, using a luni-solar calendar.

Tritya Brahmana the words sun, moon, planets, samvat (year), seasons, months, half months, arhotra (day and night), purnima (full moon) etc appear. I

In Tritiya Sanhita there is a mention of the names of the seasons as well as the months.

The Jyotisha Vedanga, compiled between 5th-4th century BCE, discusses various aspects of astronomy. It mentions the 5-year luni-solar cycle or yoga, with 67 sidereal months, 62 synodic months, and 1830 solar days.

Each solar year has 366 days, six seasons, and 12 solar months. Additionally, one month has 30 lunar days, known as tithi. This was used to calculate the day and time for the performance of rituals and sacrifices. The days are further divided into smaller units of time such as muhursats, nadika, Ica/a, etc. This method of calculating the time for rituals is still in use.

Siddhantas Another set of astronomical literature, known as Siddhantas, written around 400 CE, detailed various astronomical concepts in the Indian subcontinent.

There were 18 Siddhantas, out of which the Surya-Siddhanta, is the most well-known.

The Siddhantas, compared to the earlier works, involved greater application of mathematics and computations in astronomical concepts such as

- the motion of planets;
- their mean and true positions;
- their synodic periods;
- the orbits of various astronomical bodies;
- the position of the moon and the sun; and
- solar and lunar eclipses

Kerala School of Astronomy and Mathematics The text also mentions the nakshatras which are During the 14th-16th centuries, the Kerala School of Astronomy and Mathematics developed in the Indian subcontinent, where various astronomers tried to solve practical problems of astronomy.

Mughal Empire. The Mughal Emperors commissioned several astronomical texts and treatises such as Zij- i ilahi and Zij-i-Muhammad Shahi, which compiled astronomical data and theories, serving as reference works for scholars and astronomers. They also patronized scholars like Mir Fathullah Shirazi, who translated Sanskrit astronomical texts into Persian.

Mughal astronomers developed sophisticated devices for observation and measurement. These instruments, including astrolabes, sundials, and celestial globes, enabled precise calculations and observations of celestial phenomena.

Maharaja Sawai Jai Singh II The culmination of the Indian tradition of astronomical observation through instruments occurred in the 18th century. Maharaja Sawai Jai Singh II of Jaipur, contributed to the major development of the school of astronomy in India.

The Maharaja was a great scholar and astronomer and was well versed in various European and Arabic astronomical works. He patronised the translation of Ptolemy's Almagest and Euclid's Elements, into Sanskrit by his court astronomer, Jagannatha.

British rule In the 18th-19th centuries, with the advent of British rule in the Indian subcontinent, further developments took place in the field of astronomy.

During the colonial period, the adoption of modern technologies and methodologies bolstered India's capabilities in exploring the cosmos.

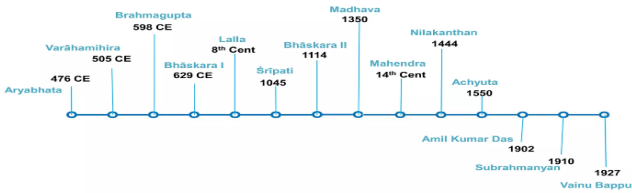
Instruments like the astrolabe, used for measuring celestial positions, and the telescope, instrumental in observing distant objects, were integral to India's astronomical pursuits.

Use of Astronomy;

1. Determining time/ time keeping
2. Tracking seasons,
3. Agriculture planning - to understand when the best times were to plant crops leading to more efficient planting and harvesting
4. Navigation-Direction determination
5. creation of maps and allowed people to know where they were going and how to get back home,
6. create astrological charts and read omens(a sign of something that will happen in the future)
7. Developing sophisticated mathematical models and many intriguing theories,
8. Calendar
9. Astrology-Foretelling the future/forecasting of horoscopes.
10. Trade between distant nations to take place.
11. Religious functions
12. Architecture and urban planning- alignment, vastu
13. Eclipse and planetary movement

INDIAN ASTRONOMERS

S.NO.	ASTRONOMER	YEAR
1	Aryabhata	476–550 CE
2	Varāhamihira	505 CE
3	Brahmagupta	598–668 CE
4	Bhāskara I	629 CE
5	Lalla	8th century CE
6	Śrīpati	1045 CE
7	Bhāskara II	1114 CE
8	Madhava of Sangamagrama	1350–1425 CE
9	Mahendra Suri	14th century CE
10	Nilakanthan Somayaji	1444–1544 CE
11	Achyuta Pisārati	1550–1621 CE
12	Amil Kumar Das	1902–1961
13	Subrahmanyan Chandrasekhar	1910–1995
14	Vainu Bappu	1927–1982



IMPORTANT ASTRONOMERS

TIMELINE

Aryabhata (476-550 CE)

1. The first person to make significant contribution to it was Aryabhata (476 AD – 550 AD) who wrote a book titled Aryabhatiyam.
2. Aryabhata estimated the circumference of earth and his estimation was very close to modern calculations.
3. He postulated a new theory that the earth was round and that it was rotating on its own axis.
4. Aryabhata was also the first to explain the true causes of solar and lunar eclipses.
5. He said eclipses were caused by relative motion of sun, moon and earth and that shadows of moon and earth were responsible for eclipses.
6. Aryabhata also put rules of planetary movement.
7. Heliocentric (somewhat) idea :- referred to or measured from the sun's center or appearing as if seen from it. 1000 years before Copernicus and Galileo
8. Described methods of eclipse prediction.
9. Knew of most of the planets (at least 7) without a telescope
10. Came up with celestial calculations.
11. Calculated earth rotation per solar orbit, days per solar orbit, and days per lunar orbit.
12. Very Accurate information without early century sources.
13. Also calculated value of pi and solar year.

1. The Aryabhatiya: The first person to make significant contribution to it was Aryabhata (476 AD – 550 AD) who wrote a book titled Aryabhatiyam. This treatise contains specialized sections on spherical astronomy (Gola) and the measurement of time (Kalakriyapada).
2. Circumference: Aryabhata estimated the circumference of earth and his estimation was very close to modern calculations. He estimated the Earth's circumference to be 39,968 km.
3. Earth's Rotation and Motion: He asserted that the earth is spherical and rotates on its own axis from west to east, explaining the apparent daily motion of stars as a relative phenomenon.
4. Scientific Explanation of Eclipses- Aryabhata was also the first to explain the true causes of solar and lunar eclipses.
5. Reflected Sunlight: He accurately stated that the Moon and planets do not have their own light but shine by reflecting sunlight. He said eclipses were caused by relative motion of sun, moon and earth and that shadows of moon and earth were responsible for eclipses.
6. Heliocentric (somewhat) idea :- referred to or measured from the sun's center or appearing as if seen from it. 1000 years before Copernicus and Galileo
7. Sidereal Year: He calculated the length of the sidereal year to be 365.25858 days, a very close approximation to the modern value.
8. Sidereal Rotation: He calculated the earth's rotation to be 23 hours, 56 minutes, and 4.1 seconds.
9. Knew of most of the planets (at least 7) without a telescope

• **Brahmagupta (598 – 668 CE)** Brahmagupta (598 AD – 668 AD) was another great astronomer in India

Wrote astronomical text, Brahmasphutasiddhanta (628 CE). Consisted of 25 Chapter; explanation of astronomical and mathematics concepts.

Covered various topics such as conjunction of the planets and problems of diurnal rotation.

1. He wrote Brahmasphutasiddhanta and Khandakhadyak. He calculated the instantaneous motion of a planet, gave correct equations for parallax, and some information related to the computation of eclipses.
2. Solar year He notably calculated the, solar year as 365 days, 6 hours, 12 minutes, and 9 seconds.
3. Eclipse Prediction: Provided accurate, techniques for calculating the timing and duration of solar and lunar eclipses.
4. Earth as a Sphere: Disputed the Puranic, belief of a flat Earth, proving that the Earth is a sphere, and estimating its circumference to be around 36,000 km. Surprisingly accurate! (off by less than 10%)
5. Planetary Motion and Positions: Developed advanced methods for calculating the mean and true longitudes of planets
6. Gravity Concept: Argued that bodies fall towards the Earth because it is in the nature of the Earth to attract objects, using the term *gurutvākarṣaṇam*. 2351 years before Newton proposed the Law of Gravitation in 1687 AD!)
7. Lunar Illumination: Explained that the Moon's brightness depends on its relative position to the Sun.
8. Astronomical Instruments: Discussed various instruments used for observing, the sky, and calculating the, rising and setting of celestial bodies.

Varahamihira (505 – 587 CE)

- Varahamihira (505 AD – 587 AD) was also a great astronomer in India. Varahamihira's Panchasiddhantika (meaning "Five Treatises on Astronomy") is a comprehensive compilation and comparison of five earlier astronomical systems or siddhantas
1. Described the Earth as spherical and as being suspended in space
 2. Used latitude and longitude
 3. Described the stars as "fixed" and gave the planets a constant speed.
 4. Brihat Samhita (Encyclopedic Work): Covers a broad range of scientific topics, including astronomical observations, planetary motions, eclipses, timekeeping (sundials and water clocks), and cloud formation.
 5. Planetary Motion and Eclipses: Improved methods for calculating the positions of planets and predicting eclipses.

6. Brihat Jataka: A fundamental treatise on horoscopic astrology that refined calculations and zodiac studies. ,
7. Precession of the Equinoxes: Varahamihira was the first to identify that the ayana (equinox) shifts by 50.32 arc seconds per year.
8. Long before Newton, this was an advancement on Anaximander's concept of equilibrium and recognition of a proto-gravitational theory.
9. Varahamihira proposed that objects must be held stationary by an attractive force.

BHASKARA I (7th century CE) Bhaskara I (c. 600–680 CE) was a pivotal 7th-century Indian mathematician and astronomer who advanced Aryabhata's astronomical school

1. Mahābhāskarīya ("Great Book of Bhaskara"): An eight-chapter work covering planetary longitudes, heliacal risings and settings, conjunctions of planets and stars, lunar crescents, and solar/lunar eclipses
2. Āryabhaṭīyabhāṣya (629 CE): A profound commentary on the Aryabhatiya of Aryabhata I, which is among the oldest known prose works in Sanskrit on mathematics and astronomy.
3. Planetary and Eclipse Calculations: He improved methods for calculating the true positions (longitudes) of planets and the timing of solar and lunar eclipses.
4. Lunar Crescent and Visibility: He provided methods to calculate the phases of the moon and the visibility of the planetary crescent.
5. Heliacal Rising and Setting: He calculated the times of heliacal risings and settings of planets

BHASKARA II (1114-1185 CE), also known as Bhaskaracharya, was a leading Indian mathematician and astronomer

1. **Siddhanta Shiromani:** Written in 1150, this treatise consists of Grahaganita (mathematics of planets) and Goladhyaya (study of the sphere), detailing planetary mean motion, true motions, and syzygies.
2. **Planetary Motion & Eclipses:** Developed advanced formulas for calculating accurate planetary positions, conjunctions, and solar/lunar eclipses.
3. **Sidereal Year Length:** the time it takes for Earth to complete one full orbit around the Sun relative to the fixed stars, Accurately computed the length of the sidereal year.
4. **Astronomical Calculations:** Wrote the "Goladhyaya" (Sphere) chapter, focusing on the geometry of celestial bodies.
5. **Calculated the time for Earth's rotation on its axis and its orbit around the Sun. Eclipses and Planetary Positions:**
6. **Provided accurate calculations of solar and lunar eclipses and the positions of planets. Earth's Circumference:**
7. **Estimated the Earth's circumference with remarkable precision for his time.**
8. **Instrument Design:** Described the construction and use of astronomical instruments, including the armillary sphere.

LALLA (C. 720–790 CE) was a prominent Indian astronomer and mathematician who followed the tradition of Aryabhata I, writing in the 8th century.

1. **His work, primarily compiled in the Śiṣyadhīvrddhidatantra, focused on refining planetary motion calculations and correcting existing misconceptions in as**
2. **Correction of Planetary Parameters:** Lalla produced a set of corrections for the Moon's longitude and updated the planetary positions calculated by Aryabhata I, which were 250 years old by his time.
3. **True Longitude Calculation:** He provided detailed methods for determining the "true" positions of planets (sphuṭa) from their mean positions (madhyama).
4. **Planetary Motion:** He extensively detailed the mean and true motion of planets, including the *sighrocca* (conjunction point) of planets.
5. **Heliacal Risings and Settings:** Lalla included calculations for the invisibility of planets due to their proximity to the Sun.
6. **Eclipses:** He provided detailed, updated methods for predicting solar and lunar eclipses, including calculations for the diameter of the earth's shadow.
7. **Astronomical Instruments:** Lalla described the use of specialized instruments for observation, such as the Shalaka Yantra (needle instrument) for finding the altitude of heavenly bodies.

INDIAN CALENDAR

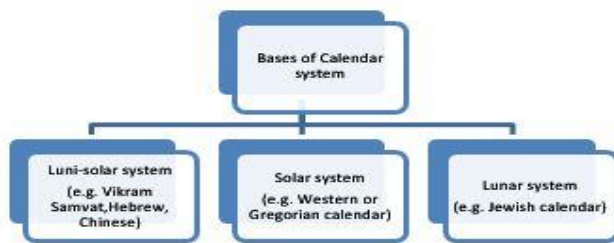
A calendar is a chart, device or piece of software which displays the date and the day of the week, and often the whole of a particular year divided up into months, weeks, and days.

calendar, any system for dividing time over extended periods, such as days, months, or years, and arranging such divisions in a definite order.

Various types of Calendar follow across the World

All types of Calendar system worldwide can be categorized, based on astronomical years which depends on movement of celestial bodies, mainly under 3 types as below.

1. Solar calendars
2. Lunar Calendar system
3. Luni-Solar Calendar system



SOLAR CALENDARS

Solar Calendar is finalized on the basis of movement of the Earth around the Sun. Solar System ignores Moon completely and it is based on Sun only

One Year in solar system is the time that earth requires to complete one orbit around the sun, which exactly is 365 days 5 hours 48 minutes and 46 seconds i.e 365.242199 days.

It has 365 days in a year but every fourth year is a leap year i.e. an extra day is added to the month of February.

Structure: A standard year has 365 days, divided into 12 months, with a leap day (February 29) added nearly every 4 years to account for the extra 1/4 day (approx.) in the Earth's orbit.

Indian Solar Calendars used in the areas of Assam, Bengal, Haryana, Kerala, Punjab, Orissa, Tamil Nadu and Tripura.

An actual solar calendar doesn't consists of number of days, hence in order to synchronize, days are summed up and leap years are formed. Indian National Calendar comes under this category Lunar Calendar system

SOLAR CALENDARS

- **Julian Calendar, Gregorian calendar, Malayalam calendar and Thai calendar are some of the examples of solar calendar.**
- **Gregorian Calendar: The international standard, refined for high accuracy.**
- **Julian Calendar: Preceded the Gregorian system, featuring a simpler 4-year leap cycle.**
- **Egyptian Calendar: One of the earliest, using 365 days based on the rising of Sirius.**
- **Solar Terms: Used in East Asian lunisolar calendars (e.g., Chinese) to track specific seasonal points.**

Purpose: Primarily designed to track seasons for agriculture, planning, and ensuring solar events (like equinoxes) fall on similar dates.

LUNAR CALENDAR

Lunar system ignores Sun completely and it is based on Moon only. It is based on Movement of the moon around the earth

The lunar calendar uses the phases of the moon to measure time, usually measuring the time from new moon to new moon as one month.

One revolution of the moon around the earth equals one month.

- It has 12 lunation or months comprising of 29 to 30 days i.e. the 12 constellations marking the path of the sun. In all the regions, each month is divided into two halves
- The dark half of the month is known as Krishnapaksha
- The Krishnapaksha begins with the new moon and ends with a full moon
- The bright half of the month is known as Shuklapaksha
- The Shuklapaksha begins with the New Moon and ends with the full moon
- In a cycle of five years, every third and fifth year has thirteen months

Months: A "lunation" or synodic month (time between new moons) is roughly 29.5 days.

Year: A 12-month lunar year lasts about 354 days.

Pure Lunar: Islamic calendar, which shifts against seasons.

Usage: Used to determine religious festivals (e.g., Ramadan, Lunar New Year).

LUNI-SOLAR CALENDAR SYSTEM

Luni Solar system takes both Sun and Moon into consideration.

By definition, a lunisolar calendar is based on the orbital motion of the Sun and the Moon as observed from the Earth (i.e. in a geocentric model).

It uses the solar year but divides it into 12 lunar months.

The definition of a lunar month is the time required for the moon to orbit once around the earth and pass through its complete cycle of phases.

At the same time this month also considers successive entrances of the sun into 12 rasis – signs of the zodiac, i.e. the 12 constellations marking the path of the sun.

Indian Calendars used in Andhra Pradesh, Bihar, Gujarat, Karnataka, Kashmir, Madhya Pradesh, Maharashtra, Rajasthan, Uttar Pradesh

Various Calendars in India

In India, basically, the four types of calendars are followed, which are as follows:

1. Vikram Samvat, also known as the Hindu lunar calendar.
2. Saka Samvat, also known as the Hindu Solar calendar.
3. Hijri Calendar, also known as the Islamic lunar calendar.

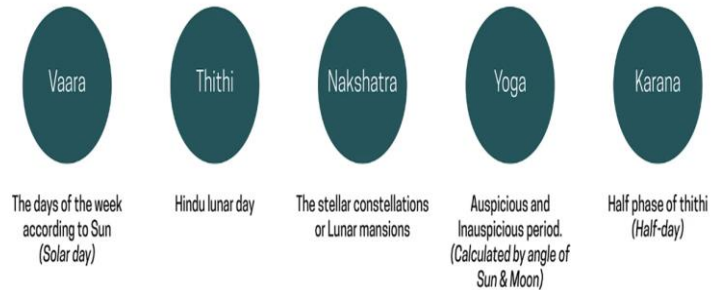
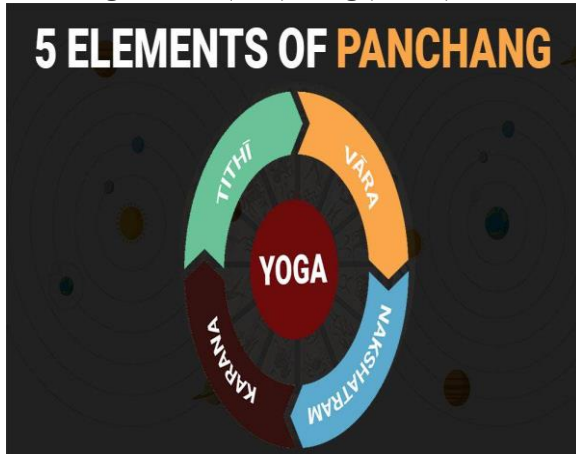
4. Gregorian Calendar, also known as the Scientific solar calendar

PANCHANG

The Indian Calendar - also called Panchang or Panchanga, Since ancient times, this Calendar has been used by people to calculate auspicious times and is truly a scientific tool.

The Panchang calendar is derived from two words - Panch means five and ang mean aspects. Together the Calendar means that it uses five aspects to function, i.e.

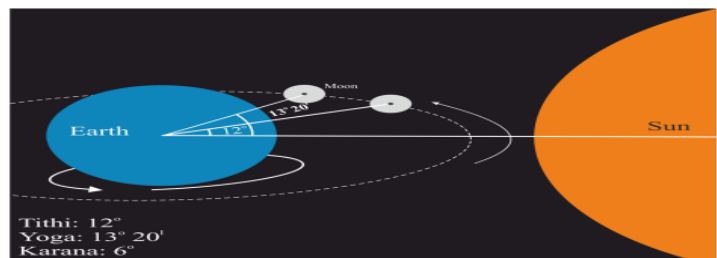
Panchang = Panch (Five)+ Ang (Limbs)



Together the Calendar means that it uses five aspects to function, i.e.

1. Vaaram/Vaar (Days)
 2. Thithi (Lunar days)
 3. Karana is half a thithi
 4. Naksatra (asterism)
 5. Yoga (Auspicious & inauspicious periods)
-
1. Var: day of the week (Sunday, Monday etc),
 2. Tithi: which serves the purpose of date,
 3. Karan: which is half of the tithi at that time.
 4. Naxatra: A division of the sky (constellation) used for determining astrological traits based on the moon's position
 5. Yoga (sum): Every day's angle of Surya (Sun) with Chandra (Moon).

Panchanga – The five limbs of Indian calendar. These five attributes depend upon the motion of the Moon.
Tithi/Date: One Tithi equals 12 degree difference between Moon and Sun.
Nakshatra/Lunar mansion: One Nakshatra equals 13 degrees:20 minutes. There are 27 Nakshatra in 360 degrees.
Yoga: One Yoga equals 13 degrees:20 minutes. There are 27 Yogas in 360 degrees.
Karana: One <i>Karana</i> equals 6 degree difference between Moon and Sun.
Vaara: The seven weekdays.



Tithi: is the time taken by the moon in increasing its distance from the sun by 12 degrees.

Yoga: The period of time during which the distance between the Sun and the Moon is increased by 13o 20' [13 degrees and 20 minutes], which is about one Day. There are 27 Yoga in 360o .

Karana: Karna is the half of tithi, i.e. Tithi is 12° increase in a distance between sun and moon and Karana is 6° increase in distance between sun and moon.

Vaaram/Vaar (Days)

The counting of days begins with the sunrise of the first day to the sunrise of the second day. In astrology, this system is known as Ahoratra, which means day and night taken together. Each day consists of 24 Horas. Vāsara refers to the weekdays and the names of the week

- Vāsara refers to the weekdays and the names of the week
- In the Indian Panchang, the days of the week are seven (as in Gregorian calendar).
- These are: Sunday, Monday, Tuesday, Wednesday, Thursday, Friday and Saturday.
- The Sun is regarded as the lord of Sunday, Moon the lord of Monday, Mars the lord of Tuesday, Mercury the lord of Wednesday, Jupiter the lord of Thursday, Venus the lord of Friday and Saturn the lord of Saturday

Days of the Week	सप्ताह के दिन	HindiTeacherOnline.Com
Monday	Somvaar	सोमवार
Tuesday	Mangalvaar	मंगलवार
Wednesday	Budhvaar	बुधवार
Thursday	Guruvaar	गुरुवार
Friday	Shukravaar	शुक्रवार
Saturday	Shanivaar	शनिवार
Sunday	Ravivaar	रविवार

Thithi & Karana (Lunar days)

- Tithi is determined by the difference between the positions (measured in Bhangash) of the Sun and moon.
- During a lunar month, moon completes one circle. Therefore this difference between the positions of these two varies from 0 to 360°.
- Tithi: is the time taken by the moon in increasing its distance from the sun by 12 degrees.
- Each Tithi is 12 degrees, there are 30 Tithis overall, this covers complete 360-degree angular movement which takes 29.5 days, this is the time taken by moon to make one revolution around the Earth
- If the difference is between 0° and 12°, it is first tithi.
- If it is between 12° and 24°, it is second tithi and it continues like this to 3rd, 4th, 5th tithi, till 14th tithi is reached.
- After that half a revolution is completed and it is either full moon or new moon.
- After that again tithis are counted from 1 to 15.
- Tithis are mentioned in two ways. According to one convention, these start with new moon day and tithi is counted from one to thirty.
- According to an alternative convention, these are counted from 1 to 15 for each half of lunar month called (Paksh).
- From new moon to full moon called (Shukal Paksh) and from full moon to new moon called (Krishan Paksh).

a Panchang month has fifteen Tithis or dates in each Paksha.

1. Prathama.
2. Dwitiya.
3. Tritiya.
4. Chaturthi.
5. Panchami.
6. Shashthi.
7. Saptami.
8. Ashtami.
9. Navami.
10. Dashami.
11. Ekadashi.
12. Dwadashi.
13. Trayodashi.
14. Chaturdashi.
15. Poornima or Amavasya depending on the Paksha.

Hindu Calendar Month

The Hindu calendar, also called Panchanga.

1. Chaitra : March/April
2. Vaisakha : April/May
3. Jyeshtha : May/June
4. Ashada : June/July
5. Shraavana : July/August
6. Bhadra : August/September
7. Ashwina : September/October
8. Kartika : October/November
9. Agrahayana : November/December
10. Pausa : December/January
11. Magha : January/February
12. Phalguna : February/March

The lunar month is exactly 29 days 12 hours 44 minutes and 3 seconds long and twelve such months create a lunar year of 354 days 8 hours 48 minutes and 36 seconds

For calculation of length of a month, Indian calendar system uses the interval between two consecutive new moons (no moon) or full moons as a basis.

According to Panchang (which is sort of Hindu Almanac) every month has 30 tithis which are equally divided between 2 pakshas.

Nakshatra' originates from 'Naksha' meaning map and 'Tara' meaning star.

- Nakshatra is a constellation or lunar mansion that represents a specific segment of the sky.
- This 27-day cycle has been taken to mean a particular group of stars.
- This has to do with the periodicity with which the Moon travels past the specific star fields called nakshatras..
- 'Nakshatra' in Indian astrology refers to constellations or groups of stars with the moon at a specific angle that affect a person's life from the moment of birth
- Nakshatras, which are variously called lunar houses or asterisms. These reflect the moon's cycle against the fixed stars, 27 days and 7¼ hours, the fractional part being compensated by an inter-calary 28th nakshatra

NAKSHATRA			
ASHWINI	PUSHYA	CHITRA	UTTARA ASHADHA
BHARANI	BHARANI	SWATI	SHRAVANA
KRITTIKA	ASHLESHA	VISHAKHA	DHANISTHA
ROHINI	MAGHA	ANURADHA	SHATABHISHA
MRIGASHIRSHA	PURVA PHALGUNI	JYESHTA	PURVA BHADRAPADA
ARDRA	UTTARA PHALGUNI	MULA	UTTARA BHADRAPADA
PUNARVASU	HASTA	PURVA ASHADHA	REVATHI

Elements of Indian Calendar

4

Yoga

"yoga" refers to specific combinations of planetary positions and relationships that influence a person's life and characteristics.

Types of Yoga:

- Planetary Combinations.
- Nakshatra Yoga.
- Karmic Yoga.

- The word Yoga translates to "union" or "addition," representing the combined celestial energies of the Sun and the Moon at a specific time.
- There are 27 distinct Yogas, each lasting for a duration of 13°20' (the same span as a Nakshatra). They are calculated by adding the celestial longitudes of the Sun and Moon and dividing the sum into 27 equal parts.
- Influence: Yogas directly impact the overall quality of the day, helping to identify the best times for significant activities

$\text{Yoga} = (\text{Longitude of Sun} + \text{Longitude of Moon}) / 13^{\circ}20'$

USE OF CALENDAR

used to determine

1. Time
2. Day
3. Year
4. seasons
5. weather
6. Dates of festivals,
7. Religious ceremonies,
8. Muhurat timing
9. The zodiac signs
10. Grow crops
11. Harvesting time
12. Planning rituals
13. Guiding personal and professional decisions
14. Astrological analysis and remedies
15. Aligning with nature's cycles

PUBLIC ADMINISTRATION AND GOVERNANCE;

The Vedic period or Vedic age (c. 1500 – c. 500 BCE) gets its name from the Vedas. Early Vedic Aryans were organized into tribes rather than kingdoms. The chief of a tribe was called 'Rajan.' The main responsibility of the Rajan was to protect the tribe. He was aided by several functionaries, including the purohita (chaplain), the senani (army chief), dutas (envoys), and spash (spies)

The earliest evidence of ancient Indian administration can be found in the Indus Valley Civilization

The administrative system in India has a rich history that dates back to the Indus Valley Civilization, which existed over 5000 years ago. During this time, the king held immense power and was responsible for overseeing all aspects of the kingdom.

He was supported by a council of ministers, officers, and other functionaries, which meant that the administration was heavily centralized around the institution of the king.

Following this period was the Vedic era, during which the early Vedic Aryans were organized into tribes rather than kingdoms. The administrative systems became more structured and organized with the rise of the Mauryan and Gupta dynasties. Both of these dynasties implemented sophisticated governmental machinery that facilitated the efficient execution of state functions. The Rig Veda period's administrative units were kull, gramme, vesh, and nation. The Vedic king's rule was aided by the samiti and sabha, two popular bodies. When people gather in a samiti, it is called an assembly. According to tradition, the king was required to attend the samiti. The sabha also served as the country's court of justice. According to the Vedas, national life and activities were first expressed through public assemblies and institutions. The king's primary duty was to safeguard his subjects. Under this administration, Gramini was in charge of the village and Senani was in charge of the army. The purohit was the most important of the many ministers who served under the king.

The Post- Vedic Period: There were many kingdoms and Mahajanpadas during the Buddha's time. Magadha, Avanti, Vats, and Kaushal were the four major kingdoms, in addition to there publics

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So the structure was King & Kingdom – There is no mention of Democracy

Key Features of Ancient Indian Administration

1. **The King and Council:** The king was central to the administration, but was advised by a Mantriparishad (council of ministers) and assisted by officials like the Purohit (priest) and Senani (army commander).
2. **Administrative Divisions:** Large empires (like the Mauryan or Gupta) were divided into provinces (Bhukti or Pradesha), which were further subdivided into districts and villages (Gram).
3. **Centralized Bureaucracy:** The Mauryan era featured a specialized, hierarchical bureaucracy that managed departments for revenue, mining, and, according to scribd.com, a vast spy network.
4. **Village Governance:** The village was the basic unit, often self-sufficient and managed by a village headman (Gramika or Gram-pati) and village councils (Sabha, Samiti).
5. **Revenue System:** Land tax was the primary income, generally ranging from 1/6 to 1/3 of the produce, though it could be as high as 1/2 in certain periods, says India Foundation.
6. **Judicial System:** The king served as the highest court of appeal, often advised by priests and elders, rather than relying on written laws.

State Administration includes -

- (a) Land Records
- (b) Law and Order including classification of crime, courts & punishment
- (c) Salary for Employees,
- (d) Hospitals and Medical Support for Citizens
- (e) Water management,
- (f) Transport – Horse, Bullock Carts, Handyman and palanquin

Governance:

King = Svamin ;

Capital = Durg = Citizens lived (inside & adjoining)

Prime Minister for the Kingdom = Maha Mantri

Cabinet Ministers = Mantri Parishad

Minister = Amatya for every Janapada (Territory)

Finance Minister = Kosha Adyaksh

Defence Minister = Danda Mantri

Law & Order :

Courts were classified into:

Criminal Courts & Civil Courts

There were different level of courts

District Criminal Courts;

State Criminal Courts

These courts had respective officers

Superintendents;

Chancellors,

Judges & Jury

King

Mauryan Administrative System

The Mauryan Empire marked a significant period in Indian history, as it introduced a new era of political and administrative organization.

Levels of Administration

The empire used a hierarchical structure to manage its vast territory:

Central: Led by the King, assisted by the Mantriparishad (Council of Ministers). High officials called Tirthas (18 in total) managed major departments.

Provincial: Divided into five major provinces:

Uttarapatha (Taxila)

Avantiratha (Ujjain)

Dakshinapatha (Suvarnagiri)

Prachyapatha (Tosali)

Magadha (Pataliputra - Central Capital).

These were typically governed by royal princes called Kumaras or Aryaputras.

District: Managed by Rajukas (revenue and justice) and Pradeshikas.

Local/Village: The smallest unit, led by a Gramika (village headman) and supervised by a Gopa (responsible for 10–15 villages)

- The Mauryan administrative system was well-organized and comprehensive. It included various levels of administration, such as central, provincial, local, revenue, judicial, and military. Each of these administrative structures played a crucial role in the governance of the empire.

1. Central Administration
2. Provincial Administration
3. Local Administration
4. Revenue Administration
5. Judicial Administration
6. Military Administration

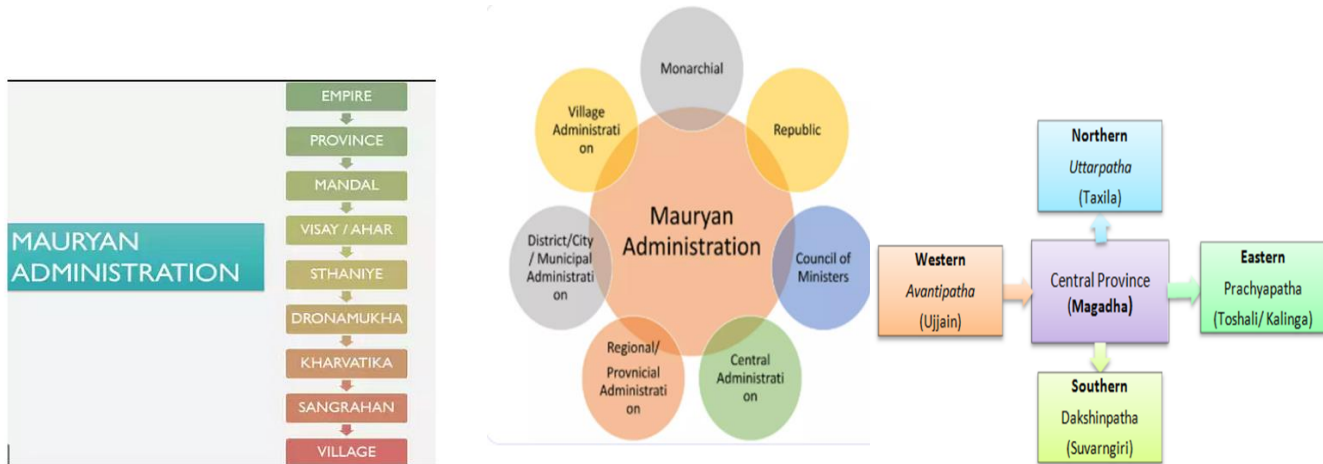


CHART OF MAURYAN ADMINISTRATION

It was a highly centralized administration

Central Administration	• King was the central figure • Next to him was a council of ministers called 'Mantri Parishad' • 2 kinds of ministers in Mantri Parishad: (i) Mantrins (Cabinet Ministers) – consisted Prime Minister, Purohita, Senapati and Yuvraja (ii) Amatyas (Bureaucrats/Civil Servants) – looked after the day-to-day administration • In addition to these ministers, the Mantri Parishad also included the leaders of 'Pura Janpatti' which was an assembly of the people of the town
Provincial Administration	• The Mauryan empire was divided into 4 provinces: (i) Northern province with capital Taxila (ii) Western province with capital Ujjain (iii) Southern province with capital Suvarnagiri (iv) Eastern province with capital Toshali • Home/Central Province was Pataliputra • They were governed by Governors called Kumar Mahamatras or Pradeshikas
District Administration	• Provinces were sub-divided into districts • 1 district – comprised about 100 villages • Officer in-charge of district – Sthanika • 1 Tehsil – comprised 5-10 villages • Officer in-charge of tehsil – Gopa
Village Administration	• It was the smallest unit of administration • A village was headed by a Gramika
Municipal Administration	• According to Megasthenes, Pataliputra was governed by a Municipal Commission of 30 members divided into 6 boards of 5 members each: (i) Industries (ii) Foreigners (iii) Registration of births & deaths (iv) Trade (v) Manufacture & Sale of goods (vi) Collection of sales tax
Judicial Administration	• King was the judicial head • Penal code was severe and harsh punishments were given • Kautilya refers to 2 kinds of courts: (i) Dharmasteya (Civil court) – Dealt with matters related to marriage, contracts, inheritance, etc. (ii) Katakashodhana (Criminal court) – Dealt with matters related to theft, robbery, murder, etc.
Revenue Administration	• 'Samharta' was the chief revenue official • A variety of taxes were collected by the state, like-land revenue, toll tax, sales tax, irrigation tax, etc. • A unique tax was also imposed on an army which passed through their territory

Military Administration	• There was a large, efficient and well-equipped army under the control of 'Senapati' • Kautilya mentions about an army called 'Chaturangha' which had 4 wings: (i) Infantry (ii) Elephantry (iii) Cavalry (iv) Chariots • Megasthenes mentioned the 5th and 6th wings of the Mauryan army: (i) Navy (ii) Transport & Commissariat
Espionage System	• Kautilya put great emphasis on the espionage system • The emperor was able to keep a watch over the remotest area of the empire and kept himself informed with all the State activities through these spies (guptaduta).

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central administration. it was directly overseen by the king. This level of administration was responsible for implementing the king's policies and ensuring the smooth functioning of the empire.

- He was responsible for the kingdom's safety and security, and he determined the overall policies to be followed by all officials. The King appointed ministers and other officers of the royal administration and served as the supreme commander of the army and head of the entire military.

The provincial administration was responsible for governing the four provinces and maintaining order within their territories.

- The provincial administration worked on similar lines of the central administration.
- The provincial Governors were responsible for day-to-day conduct of administration of provinces. They were expected to consult on important matters. (the central administration). There were also the district officers, reporters, clerks, who helped in the smooth running of provincial administration.

The local administration managed the affairs of the towns and villages within each province.

- This included overseeing public works projects, maintaining law and order, and collecting taxes.
- The revenue administration was responsible for the collection of taxes and other financial matters.
- This branch ensured that the empire's finances were in order and that funds were appropriately allocated.

The district administration was in the charge of 'Rajukas', whose position and functions are similar to today's district collectors.

- He was assisted by 'Yuktas' or subordinate officials. In the urban areas, there was a Municipal Board with 30 members.
- There were six committees with five Board members in each to manage the administration of cities.

Village administration was in the hands of 'Gramani' and his superior was called 'Gopa' who was in charge of ten to fifteen villages. Census was a regular activity and the village officials were to number the people along with other details such as their castes and occupations.

The judicial administration dealt with legal matters and the enforcement of laws. It ensured that justice was served and that disputes were resolved in a fair and timely manner.

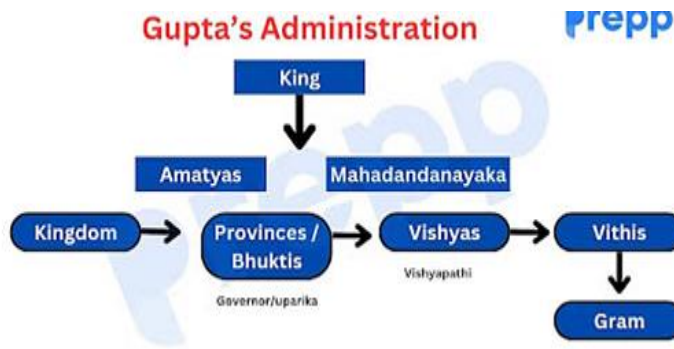
- The King served as the head of the judiciary, he appointed judges to handle regular cases.

The military administration was responsible for the defense and security of the empire.

- This branch was responsible for maintaining a strong and well-equipped army to protect the empire's borders and maintain internal stability.
- The King was the supreme commander of the military.
- The Mauryan army was well organized and it was under the control of a 'Senapati':
- six committees with five members in each. These committees were responsible to manage the following wings of the military:
- Navy
- Transport and Supply
- Infantry
- Cavalry
- War Chariots
- War Elephants
- Each of the above wings was under the control of Adhyakshai or Superintendents.

ADMINISTRATIVE SYSTEM DURING GUPTA PERIOD

- The administrative system during the Gupta dynasty was found more or less similar to that of the Mauryan Empire.
- The Empire was classified into administrative divisions like Rajya, Rashtra, Desha, and Mandala.
- This denotes the importance being given to administrative decentralization.
- Central Administration
- During the Gupta Age, the government was a benevolent monarchy, with the king holding the highest authority and possessing extensive powers for the smooth functioning of the empire.
- The king was assisted by a council of ministers, known as the Mantri Parishad, which included the Prime Minister (Mantri Mukhya) and other officials responsible for various areas such as military affairs, law and order, and more.
- These officials were known as Mahasandhi Vighraha, Amatya, Mahabaladhikari, and mahadandnyak



Administrative System during Gupta Period

- 1 Central Administration
- 2 Provincial Administration
- 3 Local Administration
- 4 Revenue Administration
- 5 Judicial Administration
- 6 Military Administration
- 7 Trade and Business

Provincial Administration

- The Guptas organized a system of provincial and local administration.
- The Empire was divided into divisions called 'bhuktis' and each Bhukti was placed under the charge of an 'Uparika'.
- The Bhuktis were divided into districts or Vishayas and each Vishaya was under a Vishayapati.
- The Vishayapatis were generally members of royal family. They were assisted in the work by a council of representatives.

Local Administration

- The local administration of the city was managed by a Parishad, with its leader referred to as Nagara Rakshaka. Another officer, called the Purapala Uparika, supervised the Nagara Rakshaka.
- In addition, there was a special officer known as Avasthika, who served as the Superintendent of Dharamsalas (rest houses).
- The village represented the smallest administrative unit. The village head, known as Gramika, was supported by other officials, such as Dutas (messengers), headmen, and Kartri.

Revenue Administration

- Revenue administration played a crucial role in the functioning of the Gupta Empire, with various officials responsible for its proper management.
- These officials included Vinayuktaka, Rajuka, Uparika, and Dashparadhika, among others.
- Their primary responsibility was to ensure the collection of taxes and other forms of revenue for the empire.

Judicial Administration

- Judicial system was far more developed under the Guptas than in earlier times.
- It was the duty of the King to uphold the law and deal with legal cases with the help of Brahman priests, judges, and ministers.
- King was the highest court of appeal.
- At the lowest level of the judicial system was the village assembly or trade guild. • These were the village councils, which were appointed to settle the disputes between the parties that appeared before them.

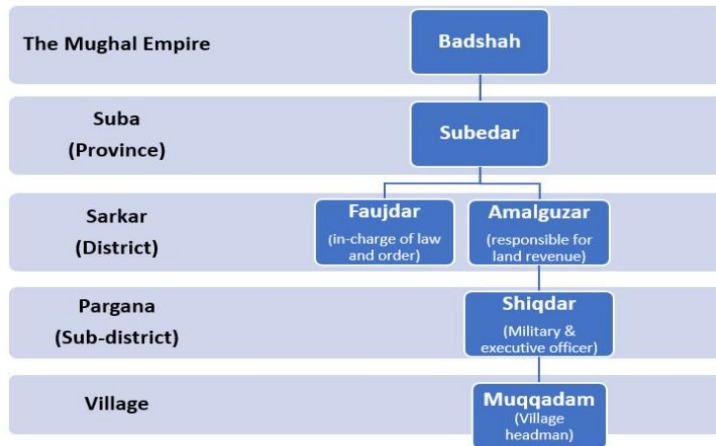
Military Administration

- The Gupta Empire was known for its powerful and expansive military, which included a standing army.
- The empire's territories were closely monitored and guarded by military officers, who held titles such as Senapati, Mahasenapati, Baladhikrita, Mahabaladhikrita, Dandanayaka, Sandhivigrahika, and Mahasandhivigrahika.
- These officers played crucial roles in managing the military administration.
- The Gupta military was organized into four main divisions:
 1. the intelligence wing,
 2. the cavalry wing,
 3. the elephant wing, and
 4. the navy.

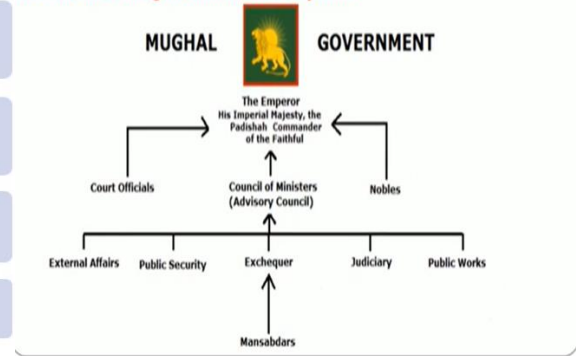
The Administration of Mughals

- The Mughal empire was famous and kept as a benchmark in the field of administration which they adopted from their ancestors.
- They exhibited a proper structure in administration which was split into three layers.
 1. Central Administration.

2. Provincial Administration.
3. Local Administration



Structure of the Mughal Administrative System



Central Administration of Mughals:

- The Emperor was the central authority in Mughal Administration where he holds the supreme power over the law making and Final arbitrator of Justice.
- He also appointed separate officers to look after the affairs of the departments allocated to them.
- The Mughal administration under the emperor had four main divisions.
- Wakil: He is the Prime Minister of the main provinces of the Mughal empire who overlooked the administration of other departments. He was under the direct control of the Emperor.
- Diwan: He is the Chief Minister who worked under the Wakil and the controller of revenue and finance of all the departments functioning in the empire.
- He also assigned the duties and revenue for Faujdars of the Provinc
- Mir Bakshi:** He was the head of **Military affairs** under the Mughal administration who looked over the recruitment of soldiers, entering their role in the **Mansabdari** system, and maintaining their ranks.
- Mir Saman:** Also known as the **Khan-i-Samman** look after the royal household. He maintained the records of “**Kharkhanas or royal workshops**”.
- Sadr:** He was the head of **religious departments**, collected charity and other revenue to enhance education in Madras

The Mughal Provincial Administration:

- The Mughal emperors divided their provinces named “Subas” which was headed by “Subedars”.
- The Suba’s are further divided into three groups.
 1. The Subas are further divided into Sarkars or districts,
 2. and when the group of villages are combined they are called Parganas.
 3. Villages or Grama are the lowest unit of administration under Mughals.

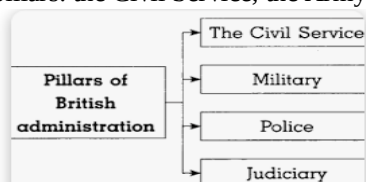
Local Administration:

- The small towns and cities are administered by Kotwals, who are the law officers of the districts and Faujdar was the head of the Sarkar.

The Village heads are known as “Muquaddams” who dealt with the local administration of villages

BRITISH ADMINISTRATION

The British Raj was the period of British rule over the Indian subcontinent from 1858 to 1947,. The British Raj was based on three pillars: the Civil Service, the Army, and the Police:



Civil services:

- The **Civil servants** are officials who implemented the laws enacted by the British company, and aggregating the revenue were their primary duty. he British introduced the name 'Civil servant' in 1829 to distinguish the civilian heads from the military personnel under the company.

Army of the British:

The army played a stellar role in the triumph of the British company and its dominance in India. he army served as the second crucial tenet of the British government in India. It served as the primary tool for extending and defending the British Empire in Asia and Africa. It was the means by which the Indian powers were subdued. It protected the British Empire in India from foreign adversaries. It waged several wars under the three presidencies.

1. Army of **Bengal** Presidency

2. Army of **Madras** Presidency and
3. Army of **Bombay** Presidency

. Police

4. The police, another institution founded by Cornwallis, served as the third pillar of British power.
5. He established a permanent police force to uphold peace and order and removed the zamindars of their policing duties

Judicial Organization

6. Through the establishment of a hierarchy of civil and criminal courts, the British set the groundwork for a new system of delivering justice.
7. Though given a start by Warren Hastings, the system was stabilized by Cornwallis in 1793. In each district was established a Diwani Adalat, or civil court, presided over by the District Judge who belonged to the Civil Service

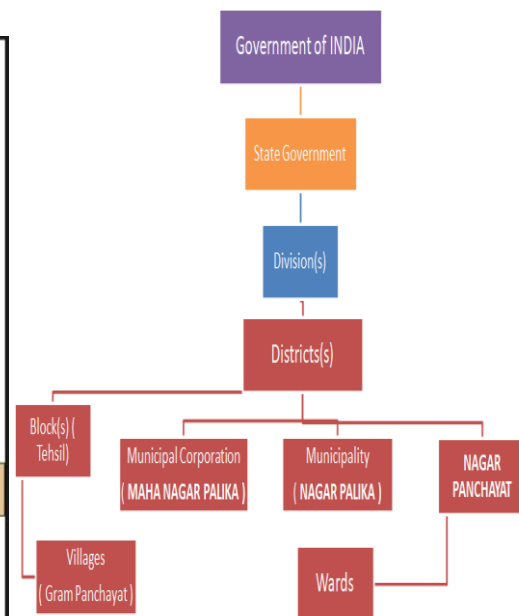
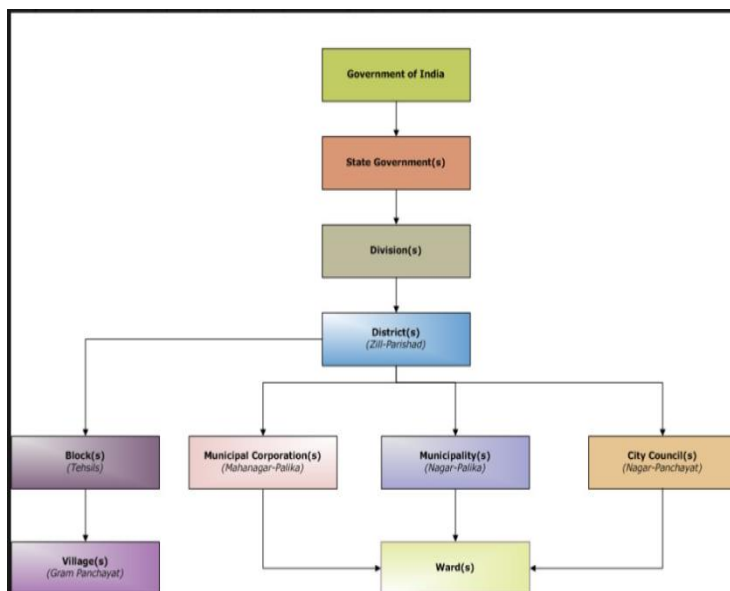
Governance & Administration present

India is a Sovereign Socialist Secular Democratic Republic with a Parliamentary form of government which is federal in structure with unitary features.

There is a Council of Ministers with the Prime Minister as its head to advise the President who is the constitutional head of the country. Similarly in states there is a Council of Ministers with the Chief Minister as its head, who advises the Governor.

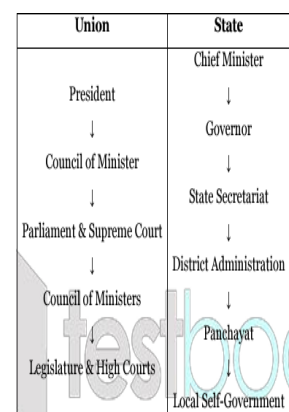
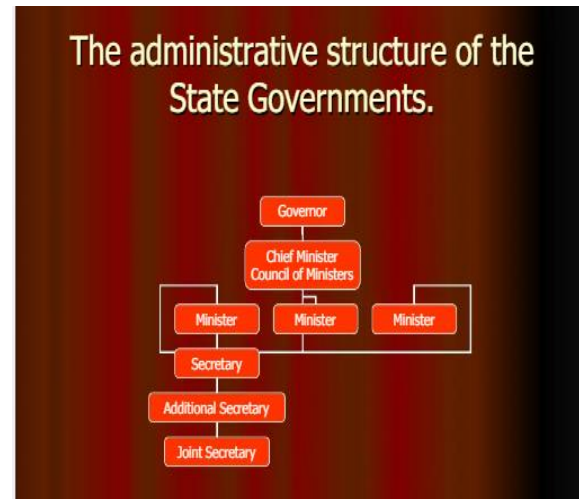
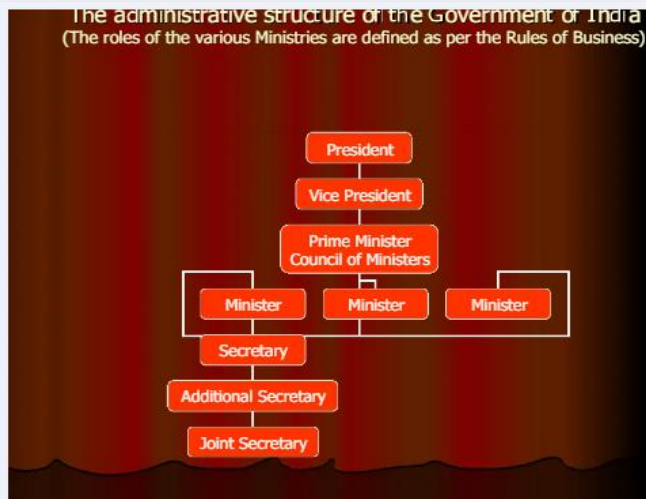
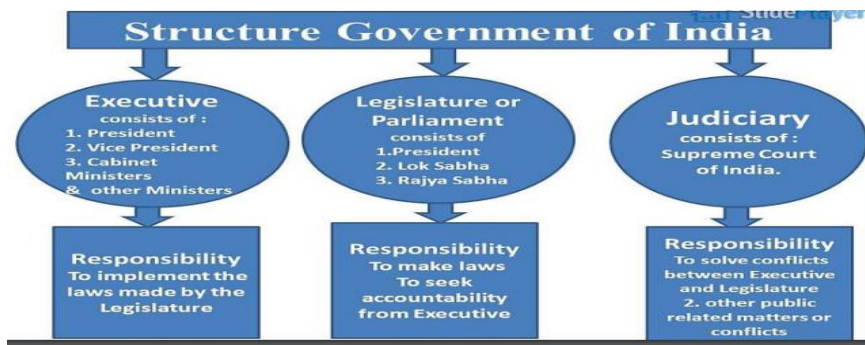
Indian governance and administration at

1. the Central,
2. state
3. local level.



Functions of Indian Administrative Systems

- ☐ Indian administration is accountable to the people of India.
- ☐ The main function is to act in accordance with objectives laid down in Preamble of the constitution, DPSP, FR.
- ☐ To secure all its citizens' liberty, equality and fraternity.
- ☐ To work for socio economic welfare of society
- ☐ To formulate and implement the projects for the economic development
- ☐ Work for government instead for the political party in power



COMPARISIO BETWEEN ANCIENT(maruan) ADMINISTRATION AND PRESENT DAY ADMINISTRATION

PARTICULAR	MARUAN ADMINISTRATION	PRESENT ADMINISTRATION
administrative system	Centralization	Centralization but less power
administrative system,	kings who claim high sounding titles and share power with Samantas, peasants, traders and Brahmanas	there are no such kings who claim high sounding titles and share power with Samantas, peasants, traders and Brahmanas
territories	Division of territory	Division of territory
administration	Bureaucracy:	Bureaucracy:
Appointment of officials	appoint officials to carry out the administration of the empire/country.	appoint officials to carry out the administration of the empire/country.
Appointment of officials	By king	central or state governments.
financing the administration	Use of revenue collection	Use of revenue collection
empire division	divided into provinces	divided into states and union territories
Protection of state and citizen	Army	Army

ECONOMY

Economy History

An economy is an area of the production, distribution and trade, as well as consumption of goods and services.

From 1st to 7th centuries India was estimated to have the largest economy in the world controlling about one third and one fourth of the world (timeline)

The country was often referred as Swarnabhumi and Swarnadweep by many writers and travelers because of its prosperity. Ancient India possessed a remarkably advanced and prosperous economy, often described as the "Golden Bird" (Sone ki Chidiya).

1. **Ancient Indian Economy**
2. **Medieval Indian Economy**
3. **British India's Economic System**
4. **Economy of Independent India**
5. **Current Indian Economy**

1. Ancient Time (776 – 300 BC)
2. Medieval Time and Islamic Golden Age (5 – 15th Century)
3. Age of Enlightenment or The Age of Reason (1770 – 1820)
4. Industrial and Economic Revolution (1820 – 1929)
5. War and Depressions (1929 – 1945)
6. Post-War Economics (1945 – 1970)
7. Contemporary Economics (1970 – present)

- **Indus Valley (3500-1800 BCE):** Urban-focused, maritime trade-oriented.
- **Vedic Period (1500-500 BCE):** Initially pastoral, shifting to settled agriculture.
- **Mahajanapadas/Mauryan (600-185 BCE):** Urbanization, standardized currency, and taxation.
- **Gupta Period (320-550 CE):** High prosperity, significant share of world GDP, active international commerce.

Key Pillars of the Ancient Indian Economy

1. **Agriculture (Krishi):** The backbone of the economy. Farmers cultivated staples like rice, wheat, and barley, alongside cash crops like cotton and indigo.
2. **Trade and Commerce:** India was a central hub in global trade networks, including the Silk Road and maritime routes in the Indian Ocean. Major exports included spices, high-quality textiles (muslin), gems, and ivory.
3. **Industry and Craftsmanship:** Skilled artisans produced pottery, metalwork (copper, bronze, iron), and, most notably, high-quality textiles.
4. **Monetary System and Finance:** While early periods relied on barter, by 600 BCE, punch-marked silver coins were introduced, becoming more standardized under the Mauryas and Guptas.
5. **State Control (Mauryan/Gupta Influence):** The Arthashastra by Kautilya highlights a structured economy with state-managed mines, forests, and tax collection (typically 1/6th of produce).
6. **Guilds (Shrenis):** Powerful professional organisations that acted as modern-day banks, trade unions, and regulatory bodies. They managed production, ensured quality, and even provided social security to their members.
7. **Metallurgy and Artisanry:** Ancient Indians were pioneers in metalwork, notably Wootz steel, which was exported to the Middle East and Europe.

Ancient times till 1707 AD	• Indus Valley Civilization (3500 BC - 1800 BC) • Economy relied on trade, facilitated by advancements in transport. • Traded terracotta pots, beads, metals, gemstones, and more. • Used ships to reach Mesopotamia for trade.
	• 6th Century BC - 3rd Century BC • Punch-marked silver coins minted during the Mahajanapadas era. • Increased trade activity and urban development.
	• Maurya Empire (321 BC - 185 BC) • Political unity and security led to a common economic system. • Increased agricultural productivity and trade. • Empire invested in building and maintaining roads throughout India. • Improved infrastructure, security, and standardized measurements boosted trade
	• 1st Century AD - 17th Century AD • India became the largest economy in the ancient and medieval world. • Controlled between 33% and 25% of the world's wealth.
Mughal Period (1526 AD - 1858 AD)	• Unprecedented prosperity under the Mughals. • India's GDP reached 25.1% of the world economy in the 16th century. • Uniform customs and tax system enforced. • Mughal Empire's GDP reached 24.3% of the world economy in 1600 AD (second largest). • Decline of the Mughals in the 18th century.
18th Century Onwards	• Rise of the Marathas and decline of the Mughal Empire. • British influence and decline of Indian industry.
British rule	• British East India Company used revenue to buy Indian goods and raw materials.

	- Colonial government used revenue for wars, not development. - India changed from exporting processed goods to exporting raw materials under British rule. - British colonial rule devastated India's economy. - India's share of world income declined from 27% to 3% under British rule.
India after independence	1950-1979 - India adopted centralized planning after independence in 1947. - Five Year Plans were implemented to develop the economy. - Investments were made in agriculture, infrastructure, and industries. - Economic growth was slow due to lack of capital, Cold War politics, and population rise. - From 1951 to 1979, the economy grew at 3.1% annually and industry grew at 4.5% annually. - Agricultural growth lagged behind industrial growth at 3.0% annually.
	1980-1990 - India's economy grew at 5.5% annually from 1980 to 1989. - Industry grew at 6.6% annually and agriculture at 3.6% annually. - Increased investment (from 19% to 25% of GDP) fueled economic growth. - India increasingly relied on foreign borrowing in the late 1980s. - This led to a balance of payments crisis in 1990, prompting economic reforms. - Liberalisation and its effects (1991 Onwards)
	- Since 1991, - India's economy grew rapidly at 6-8% annually. - GDP went from \$267 billion in 1992 to \$1.85 trillion in 2012. - India became the world's third largest economy and a major foreign direct investment destination. - India's major industries include IT, telecommunications, textiles, and others. - Agriculture is still important, with rice, wheat, and cotton as major products. - The service sector is a leading contributor to the economy. - India has a strong network of science and technology institutions. - The government aims to increase manufacturing's share of GDP to 25% by 2025. - India is expected to become the world's second largest manufacturing economy by 2017. - India is one of the largest and fastest-growing markets for food and agriculture. - India is the world's third largest producer of food. - The Indian food industry is expected to grow significantly in the coming years.

Economic Evolution:

Indus Valley (3500-1800 BCE): Urban-focused, maritime trade-oriented.

A thriving urban society with standardized weights and measures, engaging in maritime trade with Mesopotamia.

Vedic Period (1500-500 BCE): Initially pastoral, shifting to settled agriculture.

Mahajanapadas/Mauryan (600-185 BCE): Urbanization, standardized currency, and taxation.

Mauryan Empire (321–185 BCE): Introduced a highly centralized economic system and robust taxation as detailed in Kautilya's Arthashastra. India's share of world GDP reached an estimated 32%–35%

Gupta Period (320-550 CE): High prosperity, significant share of world GDP, active international commerce. Gupta "Golden Age" (4th–6th Century CE): Characterized by immense prosperity, advancements in metallurgy and textiles, and flourishing trade with the Roman world, which famously "drained" Rome of its gold to pay for Indian luxuries.

INDUS VALLEY CIVILIZATION (3500 BC - 1800 BC)

- The economy of the Indus Valley Civilization was a sophisticated, urban-based system driven by highly productive agriculture, extensive trade networks (internal and foreign), and specialized craft production.
- It was supported by standardized weights and measures, a surplus-based food economy, and thriving maritime trade with regions like Mesopotamia
- Core Economic Components:**
- Agriculture & Livestock:** The backbone of the economy, relying on wheat, barley, peas, lentils, mustard, and sesame. They were the earliest known cultivators of cotton. Domestication included cattle, sheep, goats, pigs, buffalo, and camels, used for labor and resources.
- Trade and Commerce:** A highly organized, bartering-based trade network connected to Mesopotamia (via Persian Gulf), Oman, Iran, and Afghanistan. Goods included copper, lead, gold, silver, seashells, and precious stones like lapis lazuli.
- Craft Specialization:** Urban centers produced terracotta pottery, metal tools (bronze and copper), jewelry, beads (lapis lazuli, turquoise), and ivory products.
- Infrastructure & Standardization:** The economy was facilitated by standardized, uniform weights and measures. A major dockyard at Lothal suggests a robust maritime trade, while carts were used for land transport.

ECONOMY OF MEDIEVAL (c. 8th to 18th century) was a period of immense economic prosperity, eventually becoming the world's largest economy by the 17th century under the Mughals, contributing approximately 25–33% of global GDP.

- Its economic structure was built on a vast agrarian base, sophisticated artisanal production, and extensive global trade networks.

- **Key Features of the Medieval Indian Economy:**
- **Agrarian Structure:** Agriculture was the backbone, with peasants cultivating rice, wheat, sugarcane, and indigo. Land revenue was the main state income, organized under systems like the Iqta (Delhi Sultanate) and Zamindari (Mughal), with officials (e.g., Patwari, Qanungo) managing collections.
- **Trade and Commerce:** Inland and foreign trade flourished, with India acting as a hub for international commerce. Major exports included textiles (cotton, silk), spices, indigo, and minerals. Key ports included Surat, Calicut, and Tamralipti.
- **Monetary System:** A sophisticated currency system existed, using silver tankas and later, Mughal-era rupees. Serrafs (money changers/bankers) handled credit, including hundis (bills of exchange).
- **Manufacturing and Crafts:** Skilled artisans produced goods for local markets and export, supported by both urban workshops and imperial Karkhanas (factories).
- **Urbanization:** Growth of cities like Delhi, Agra, Lahore, and Vijayanagara served as administrative, cultural, and trading hubs.
- **Economic Transitions:** The period saw a transition from a more closed early medieval economy to a more open, monetized, and export-oriented economy under the Mughals, with increasing involvement of European trader

• **MAURYA EMPIRE (321 BC - 185 BC)**

- The Mauryan Empire (c. 322–185 BCE) operated a highly organized, state-controlled, and mostly agrarian economy that united vast parts of India, creating significant prosperity through centralized administration. Political unity and security led to a common economic system.

1. Agriculture: The Economic Backbone

- Agriculture was the primary occupation and the chief source of state revenue.
Sita Lands: These were crown lands owned directly by the state and managed by the Sitadhyaksha (Superintendent of Agriculture). [2, 3]
- **Taxation (Bhaga):** Most farmers paid 1/4th to 1/6th of their produce as land tax. In special cases, like the Buddha's birthplace Lumbini, Ashoka reduced this to 1/8th.
- **Irrigation:** The state heavily invested in infrastructure, such as the famous Sudarshana Lake in Gujarat, and levied a water tax (Udakabhaga) ranging from 1/5th to 1/3rd of the crop.

2. State Control & Monopolies

- The Mauryan state exercised extensive control over key industries to ensure a filled treasury (Kosa).
- **Monopolies:** The state held exclusive rights over mining, salt production, liquor, armaments, and shipbuilding.
- **Superintendents (Adhyakshas):** Various officials regulated specific sectors, such as Panyadhyaksha (Commerce), Akaradhyaksha (Mining), and Pautavadhyaksha (Weights and Measures).

3. Trade and Commerce

- Internal and external trade flourished due to political stability and a unified currency.
- **Trade Routes:** The Royal Highway (precursor to the Grand Trunk Road) connected Taxila to Pataliputra. Major ports included Tamralipti on the east coast and Barygaza (Bharuch) on the west.
- **Exports/Imports:** India exported spices, pearls, ivory, and fine muslin textiles while importing horses, gold, and wine.
- **Guilds (Shrenis):** Artisans and merchants were organized into powerful guilds that provided technical education and acted as proto-banks.

4. Monetary System

- The empire moved toward a monetized economy using standardized coinage]
- **Pana:** The primary currency was the silver punch-marked coin, often featuring symbols like the peacock, hill, or crescent.
- **Other Coins:** Copper coins like the Masika and Kakini were used for smaller transactions.

5. Urbanization

- Economic prosperity led to the rise of major urban centers such as Pataliputra (the capital), Taxila, and Ujjain, which served as hubs for both administration and international commerce

• **GUPTA PERIOD**

- The Gupta Empire (4th–6th century CE) economy was a prosperous, primarily agrarian system characterized by thriving trade, advanced crafts, and a robust, gold-standard currency, often termed a "Golden Age"
- Agriculture, supported by improved irrigation, was the main occupation.
- Key features included specialized industries (textiles, metalwork), powerful merchant guilds, and extensive maritime trade with Southeast Asia, Rome, and Persia

1. Agriculture and Land System

- Agriculture was the backbone of the economy and the primary source of state revenue.
- **Key Crops:** Major crops included rice, wheat, barley, pulses, and cash crops like sugarcane and cotton.
- **Irrigation:** The state maintained significant irrigation works,.
- **Land Grants (Agrahara):** The period saw an increase in tax-free land grants to Brahmins and religious institutions, which expanded cultivation.
- **Taxation:** The customary royal share of agricultural produce was one-sixth (Bhaga).

2. Trade and Commerce

- **Internal and external trade flourished, reaching its peak under rulers like Chandragupta II.**
- **External Links:** India traded extensively with Southeast Asia, China, and the Byzantine Empire.
- **Exports & Imports:** Major exports included silk, spices, textiles, and ivory, while imports consisted of gold, silver, and horses from Arabia and Iran.
- **Major Ports:** Key maritime hubs included Tamralipti on the east coast and Bharuch (Broach) and Kalyan on the west coast.

3. Guild System (Shrenis)

- **Guilds were autonomous associations of artisans and merchants that played a critical role in the economy.**
- **Autonomy:** They had their own laws, which the state respected, and even maintained their own militias.
- **Functions:** Guilds acted as banks, accepting deposits and providing loans at interest.

4. Coinage and Monetisation

- **The Gupta period is renowned for its high-quality numismatic art.**
- **Gold (Dinara):** The Guptas issued the largest number of gold coins in ancient India, used for high-value transactions.
- **Silver & Copper:** Silver coins (Rupaka) were introduced primarily after the conquest of Western India for regional trade, while copper coins were scarce, suggesting that common people often used cowries for daily transactions.

5. Economic Decline

- **Toward the end of the 6th century, several factors led to economic weakening:**
- **Decline in Foreign Trade:** The Byzantine Empire learned the art of silk cultivation from China, reducing demand for Indian silk.
- **Huna Invasions:** Constant warfare with the Hunas drained the royal treasury and disrupted trade routes.
- **Debasement:** The gold content in coins declined sharply, reflecting financial strain

ARTHASASTRA

What is the meaning of “Artha” ? Process of acquiring materialistic benefit for better livelihood – without ‘Greed’

Arthashastra means the science (sastra) of wealth/ earth/polity(artha), the science of gaining wealth‘

Arthashastra means “the science of wealth” or “political economy.” It is a guide for rulers on statecraft, economic policies, military strategy, diplomacy, law, and social organization.

ARTHASASTRA is the name of the text written by widely known as “Chanakya” – son of Acharya Chanak, also known as Kautilya, Vishnugupta

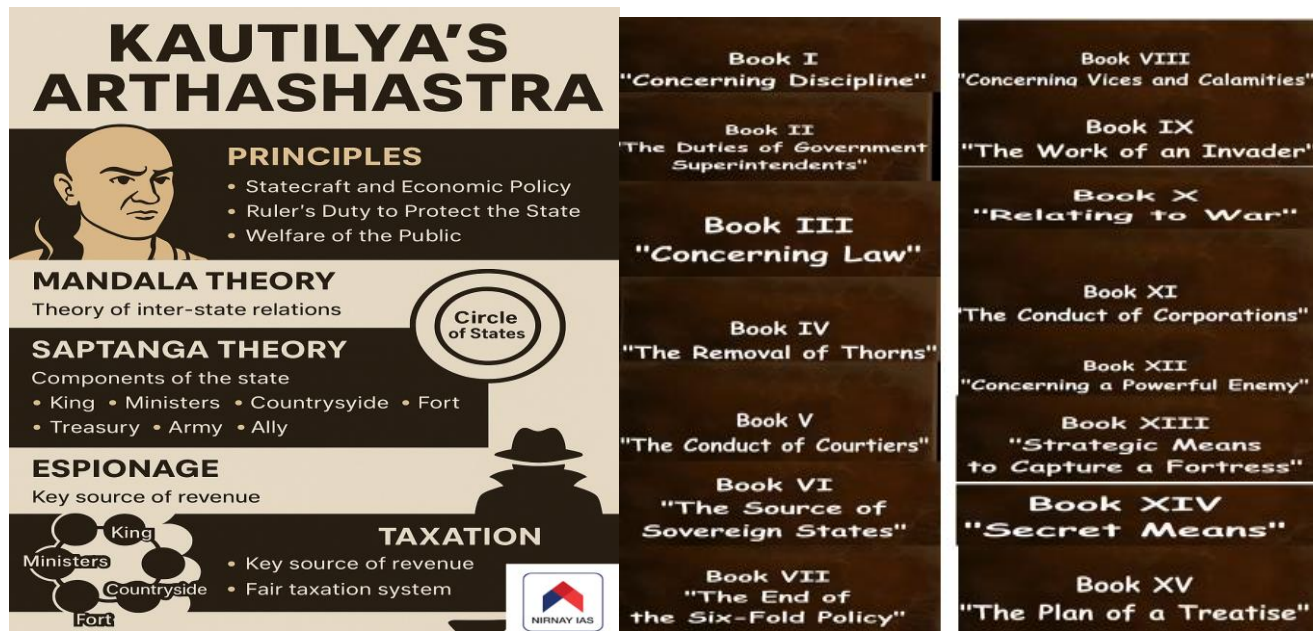
Acharya Chanakya was Guru of Chandragupta Maurya, who built the Mauryan Empire - 300 BCE

This Arthashastra text covers

- (i) State Administration
- (ii) Economic Policy
- (iii) Military Strategies :

Total of 15 volumes

- **Book 1: Concerning Discipline (Vinayadhikaranika)** – Deals with the training of the king, education, and his daily routine.
- **Book 2: The Duties of Government Superintendents (Adhyaksapracara)** – Covers the functions of various departments, administration, and economic management.
- **Book 3: Concerning Law (Dharmasthiya)** – Focuses on civil law, administration of justice, and courts.
- **Book 4: The Removal of Thorns (Kantakasodhana)** – Deals with criminal law, suppression of crime, and protecting citizens.
- **Book 5: The Conduct of Courtiers (Yogavrtta)** – Concerns the behavior of public servants, secrets of government, and strengthening the state.
- **Book 6: The Source of Sovereign States (Mandalayoni)** – Describes the seven elements of the state (Saptanga Theory) and foreign policy.
- **Book 7: The End of the Six-Fold Policy (Sadgunya)** – Discusses the six measures of foreign policy (peace, war, neutrality, etc.).
- **Book 8: Concerning Vices and Calamities (Vyasanaadhikarana)** – Analyzes calamities that can afflict the state and the king.
- **Book 9: The Work of an Invader (Abhiyatkarmasama)** – Details preparations for war and military expeditions.
- **Book 10: Relating to War (Samgramika)** – Covers battle strategies, camp arrangements, and types of warfare.
- **Book 11: The Conduct of Corporations (Sanghavrta)** – Explains how to deal with oligarchies and autonomous corporations.
- **Book 12: Concerning a Powerful Enemy (Abaliyasa)** – Strategies for a weak king to combat a stronger enemy.
- **Book 13: Strategic Means to Capture a Fortress (Durgalambhopaya)** – Focuses on conquering enemy forts and ruling conquered territories.
- **Book 14: Secret Means (Aupanisadika)** – Covers covert operations, magic, and, occult practices to overcome enemies.
- **Book 15: The Plan of a Treatise (Tantrayukti)** – Explains the methodology and techniques used in writing the book.



TRADE, The buying or selling of goods or services between people, state or countries. Trade is the act of buying and selling goods or services, often in exchange for money.

HISTORY OF TRADE AND COMMERCE

- India has thrived in trade since 2600 BC. During Harappan civilization,
- In ancient times, trade began as a barter system in which people exchanged one object for another.
- During ancient times, Indians were the masters of the seaborne trade of Europe, Asia and Africa.
- Commercial cities like Harappa and Mohenjodaro were some examples for the business development of ancient India
- These civilizations had established commercial connections with Mesopotamia and traded in gold, silver, copper, gemstones, beads, pearls, sea shells etc.
- Large extent its supplies of fine cotton and silk fabric, spices, indigo, sugar, drugs, precious stones and many curious works of art from India in exchange of gold and silver
- Portugal was the first country to establish trade relations with India
- **Trade and commerce have played a vital role in making India to evolve as a major player in the economic world in ancient times. Archaeological evidences have shown that trade and commerce was the basis of the economy of ancient India**
- **The progress of domestic and international trade in ancient India developed due to flow of wealth through these routes.**
- **Many trade centres and Industrial Belt also started. Archaeological evidences have shown that trade and commerce was the base of the economy of ancient India.**
- **The trade was mainly carried out by water and land**

TRADING COMMUNITIES

Ancient India had many trading communities, including guilds, merchants, and other groups:

Gujarati traders: Hindu Baniyas and Muslim Bohras traded with the Red Sea, Persian Gulf, East Africa, Southeast Asia, and China.

Punjabi and Multani merchants: These merchants handled business in the northern region.

Bhats: These merchants managed trade in Gujarat and Rajasthan.

Mahajan: This group was found in western India and was represented by their chief, the nagarseth, in urban centers like Ahmedabad.

Chettiars: This community was an important trading group in South India.

Manigramam and Nanadesi: These guilds in South India traded within the peninsula and with China and South West Asia.

Marwari Oswal: This community became a principal trading group of the country.

Wholesale traders: These traders were known as "seth" or "bohra".

Retail traders: These traders were known as "beoparis" or "banik".

Banjaras: This special class specialized in trading.

Trading Communities In different parts of countries, different communities dominated trade

- (1) In Northern Region Punjabi and Multani merchants handled business
- (2) In Western India Mahajans were managing trade.
- (3) From South Chettiars were important traders.
- (4) In states of Gujarat, Rajasthan Bhats managed the trade
- (5) In urban centres Mahajan Communities represented their Chief called Nagar Seth.
- (6) Other groups active in urban areas were:
 - Hakim and Vaid (physicians),
 - Wakil (Lawyer),

- Pundit or Mullas (Teachers)
- Painters,
- musicians,
- Calligrapher

MAJOR TRADE CENTRES IN ANCIENT INDIA

Patliputra: Known as Patna today. It was not only a commercial town, but also a major centre for export of stones.

Peshawar: It was an important export centre for wool and for import of horse. It had huge commercial transactions between India, China and Rome.

Taxila: It was a major centre of land route between India and Central Asia. It was also a city of financial and commercial banks. It was a place of Buddhist centre of learning. The famous Taxila university flourished here.

Indraprastha: It was a commercial junction on Royal Road. Where most routes leading to east, west, south and north converged.

Mathura: It was an emporium of trade. Many routes from South India touched Mathura and Bharuch.

Varanasi: It grew as major centre of textile industry and became famous for beautiful gold silk cloth and sandalwood workmanship. It had link with Taxila and Bharuch.

Mithila: Mithila established trading colonies in South China especially in Yunnan.

Ujjain: Agate, Carnelian, muslin and mallow cloth were exported from Ujjain to different centres.

Surat: Textile of Surat were famous for their gold border (zari). Surat's hundis were honoured in far off markets of Egypt and Iran.

Kanchi: Here Chinese used to come in foreign ships to purchase pearls, glass and raw stones and in return they sold gold and silk.

Madura: It was the capital of Pandayas. It attracted foreign merchants particularly Romans for carrying out overseas trade.

Kaveripattinam: also known as Kaveripattinam, it was scientific in its construction as city provided loading, unloading and strong facilities of merchandise. It was a centre of trade for perfumes, cosmetic, scents, silk, wool, cotton, corals, pearls, gold and precious stones.

Tamralipti: It was one of the greatest port connected both by sea and land with west and far east. It was linked by road to Banaras and Taxila.

MAJOR EXPORT AND IMPORT

1. **Major Exports included:** Species, wheat, sugar, indigo, opium, Sesame oil, cotton, parrots, Live animals and animal products like hides, skins, furs, horns, tortoise, shells, pearls, sapphire, quartz, crystals, Lapis lazuli, granites, turquoise, copper,
1. **Major Imports included:** Horses, Animal products, Chinese silk, flax, linen, wine, gold, silver, tin, Copper, , lead, rubies, coral, glass, amber, etc



TRADE ROUTE

There were two types of trade route

1. Internal
2. External

Internal

The two major trade routes of Indian subcontinent were Uttarpath of the north and north-west and Dakshinpath of the central India and the southern peninsula

1. UTTARPATH
2. AND DAKSHINPATH

The two major trade routes of Indian subcontinent were Uttarpath of the north and north-west and Dakshinpath of the central India and the southern peninsula.

UTTARPATH is also referred as "Northern High Road". This trade route included the whole of North Western India, Northern India, Eastern India and also Himalaya in the north to Vindhya in the south.

The route connected Bengal, Bihar, Gorakhpur, Bijnor, Saharanpur, Jallunder, Lahore, Taxila etc. Its Southern sector connected Lahore, Delhi, Hastinapur, Varanasi, Allahabad, Patliputra and Rajgir etc.

The Manibhadra yaksha was the presiding deity of the Uttarapath traders. This trade route was famous for its fine breed of horses and horse dealers.

DAKSHINPATH, is often referred as "Southern High Road" connected Varanasi, Ujjain, Narmada Valley and Pratisthana (Paithan — Maharashtra) and then to western coast of India, running in southern direction.

Manimekalai was the presiding deity of Dakshinpath traders.

Lapis Lazuli, silver, semiprecious stones, sea shells, gold, lead, copper, iron, sandalwood, conch-shells, diamonds, rubies, pearls, cloth and rain proof cloth were the trading articles.

These trade routes were used by merchants, tourists, pilgrims, students, actors, acrobats, kings, army, horse traders, peddlers, monks etc for their respective purposes.

These trade routes were connected to sea ports like Uttarpath which was further connected to Odisha coasts, Sind and Rajasthan.

The Dakshinpath was connected to Virampattinam (Arikamedu) through Pratisthan.

External

1. Silk route
2. The Spice Route was a way of maritime trade.
3. The Salt Route,
4. Incense Route,
5. Tin Route,
6. The Amber Road.
7. Tea route

SILK ROUTE

The Silk Road: Connected India with China via Central Asia, facilitating the trade of silk, spices, and other goods. India, the Silk Route was a series of trade routes that connected the Indian subcontinent with the rest of Asia and Europe.

These routes were important for the trade of silk, spices, precious stones, and other luxury goods that were highly valued in the ancient world.

Land route

Northern Routes:

One significant route passed through Srinagar, Gilgit, and the Karakoram Range, providing a link to Central Asia and beyond.

Central Route:

Another route went through Purushapur (Peshawar), an important city in ancient Gandhara, and connected with the main Silk Road.

Eastern Routes:

Routes through Hadda, Kapisa, and Bamiyan, located in present-day Afghanistan, were also crucial for connecting India to the broader network. Bamiyan, in particular, was a major Buddhist center and a significant stop on the Silk Road.

Maritime Routes:

In addition to the land routes, India was also connected to the Silk Road through various maritime routes across the Indian Ocean, facilitating trade with Southeast Asia and beyond.

Origin: Coined in the 19th century by German geographer Ferdinand von Richthofen, the "Silk Roads" were not a single route but a complex web of paths.

Goods Traded: While named for Chinese silk, the routes carried jade, porcelain, glassware, textiles, spices, and horses.

Cultural Exchange: It was the premier artery for the spread of ideas, technologies (compass, printing), and religions across Eurasia.

Routes: The network extended from Xi'an, China, through Central Asia, Afghanistan, and Persia to the Mediterranean. Maritime routes also connected India, Southeast Asia, and Africa.

Significance: It thrived for over 1,500 years, shaping civilizations, but declined with the rise of the Ottoman Empire in the 15th century.

Spice route

Ancient India's spice trade routes were extensive networks of land and sea routes that connected India with the Middle East, Europe, and Southeast Asia, facilitating the exchange of spices like pepper, cinnamon, and cardamom.

These routes, established around 3000 BC, were crucial for global trade and cultural exchange long before the Silk Road,

Key ports like Muziris (present-day Kerala) were vital hubs, and the trade significantly impacted the economic and political landscape of the ancient world. It is a distance of over 15,000 kilometres and, even today, is not an easy journey.

Origins and Geography: The routes began around 3000 BCE, connecting the Spice Islands in the Pacific, India, and Southeast Asia to the Middle East and the Mediterranean.

Key Commodities: While spices (pepper, cloves, nutmeg, ginger) were paramount, the route also transported, silk, incense, porcelain, and gems.

Role of India: India was the central hub, with major ports like Muziris (Kerala), Arikamedu, and Barbarikam acting as crucial trading points for exchange with Rome, Egypt, and Southeast Asia.

Key Traders: Arab traders dominated the route for thousands of years, controlling the flow of spices from the Indian Ocean to the Mediterranean.

Impact on History: The immense value of the spice trade led to the rise of trading empires and later, the Age of Discovery, as European powers sought to bypass Arab middlemen to reach India directly

The ancient Incense Route, flourishing from the 7th century BC to the 2nd century AD, was a vital network connecting the Mediterranean world (parts of Europe, Africa, and Asia) with the East, including India.

It facilitated the trade of valuable commodities like incense (frankincense and myrrh), spices, precious stones, and other luxury goods. This extensive route involved both land and sea routes, traversing through Arabia, the Levant, and Egypt, eventually reaching India and beyond.

it stretched roughly 2,400 kilometres from the production regions of southern Arabia (modern-day Yemen and Oman) and the Horn of Africa to the port of Gaza and the Levant.

The Commodities: While named for incense, the route also facilitated the trade of Indian spices, precious stones, pearls, silk, gold, ivory, and exotic animals.

Major Hubs: The network was anchored by strategic cities and ports:

Petra (Jordan): The Nabataean capital and a central transshipment point for goods heading to the Mediterranean.

Gaza (Palestine): The primary Mediterranean terminus where luxury goods were shipped to Europe.

Shabwah (Yemen): The capital of the Hadhramaut kingdom and a critical consolidation hub for frankincense.

Hegra (Saudi Arabia): A major Nabataean site (modern-day AlUla) that served as a northern customs and supply station.

Romans have developed a road for transporting amber beads from the Baltic to the east of Europe. Amber beads were used as decorative items and valued for medicinal properties. The Baltic ocean is famous for good quality amber beads which were formed millions of years ago when covered by the sea.

Amber Road are well-known in Europe, connecting the Baltic Sea to the Mediterranean, ancient India had its own extensive trade networks.

Indian Ocean Trade: Amber was likely traded along the Indian Ocean routes, which were a vital network connecting various regions.

Exchange for Goods: Amber would have been exchanged for other valuable commodities, such as precious metals, textiles, and spices.

Not a Primary Focus: While amber was traded, it wasn't the primary focus of Indian trade as it was in Europe.

Time Period: The trade flourished from the Bronze Age (around 1500 BC) through the Roman Empire, though evidence suggests amber was traded even in prehistoric Neolithic times.

Significance: Beyond trade, the routes were instrumental in cultural and technological exchanges between Northern and Southern Europe. The Roman Empire, for example, protected sections of this route due to its immense economic value.

Cultural Impact: Amber was often referred to as the "gold of the north" and was found in ancient Egyptian artifacts (such as in the tomb of Tutankhamun), testifying to the massive reach of these ancient trade networks.

TEA ROUTE

The Ancient Tea Horse Road (also known as the Southern Silk Road) was a network of trade routes connecting China, Tibet, and India, with a significant focus on tea and horses.

Tea route was developed for transporting Chinese tea and Tibetan warhorses from Hengduan mountains of China through Tibet to India.

This route was more dangerous due to terrain of crossing areas and numerous rivers.

It wasn't a single road but a complex system of paths that facilitated trade and cultural exchange for centuries.

The route originated in southwestern China (Yunnan and Sichuan provinces) and extended through Tibet, Nepal, and eventually to India, with Kolkata often serving as a final destination.

The network spanned over 2,000 to 4,000 kilometres, traversing some of the most dangerous terrain on Earth, including the Himalayas and the Hengduan Mountains.

Sichuan-Tibet Route: Started in Ya'an (near Chengdu), passing through Kangding and Batang to reach Lhasa.

Yunnan-Tibet Route: Began in the tea-rich areas of Pu'er and Xishuangbanna, moving through Dali, Lijiang, and Shangri-La toward Tibet and India.

Indian Connection: From Lhasa, the route extended through Himalayan passes to Kolkata, serving as a gateway for tea to reach European and other Asian markets.

SALT

ROUTE

The "Ancient Salt Route" refers to various historical trade networks used to transport salt.

Ancient India's salt trade routes were vital networks for distributing a crucial commodity, with significant trade occurring both inland and via coastal regions.

Key routes connected major population centers and resource areas, facilitating the exchange of salt for other goods.

These routes included both land and sea paths, with notable inland routes like the Dakshinapatha, and coastal routes linking Gujarat and other ports to the broader Indian Ocean trade network.

Economic Power: Salt was often called "white gold," and controlling its route meant immense wealth for cities and traders.

Cultural Exchange: These routes allowed not just for the movement of goods, but also for the spread of culture, technologies, and ideas across regions.

Via Salaria (Italy): One of Rome's oldest roads, it transported salt from the salt pans at Ostia to Rome and further inland to the Adriatic coast. The word "salary" originates from *salarium*, the salt allowance given to Roman soldiers.

Old Salt Route (Germany): Known as the Alte Salzstraße, this 116-km medieval route connected the salt-rich city of Lüneburg to the Baltic port of Lübeck. It was a primary source of wealth for the Hanseatic League.

Tsa'lam (Tibet/China): Also called the "Route of White Gold," this high-altitude trail crossed the Himalayan Plateau, where nomadic traders used yaks to transport salt from remote lakes to communities in need.

Salt Road (Japan): Known as Shio no Michi, these routes supplied the landlocked Shinano Province with salt and marine products from the coast.

Trans-Saharan Routes (Africa): Massive caravans, sometimes with thousands of people, transported salt from the Sahara Desert to the Mediterranean coast.